

Chapter 127

Viewers, Monitoring, and Video Scopes

Viewers let you see what you're working on, provide a UI for transport controls and image comparison, draw windows and drag on-screen controls, and provide a variety of warnings and information not found anywhere else.

However, at the same time, professional workflows require real time monitoring via supported video I/O devices on a calibrated display. Lastly, video scopes provide a more precise analysis of the state of the video signal you're viewing, and DaVinci Resolve has built-in scopes you can use for creative decision making and troubleshooting.

This chapter covers the Color page Viewer, external monitoring and display calibration, and video scopes in more detail.

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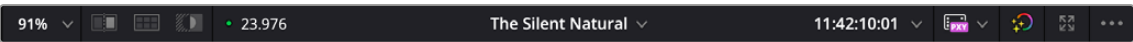
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Using the Viewer

The Viewer is your window into the Timeline. The clip and frame at the current position of the playhead appears in the Viewer. The Viewer also provides a workspace for picking colors, adjusting Windows, using split screen stills for reference, and many other display-oriented tasks. If you have a video out interface connected to a broadcast display or projector, then the contents of the Viewer are typically mirrored by the video output.

The Viewer Title Bar

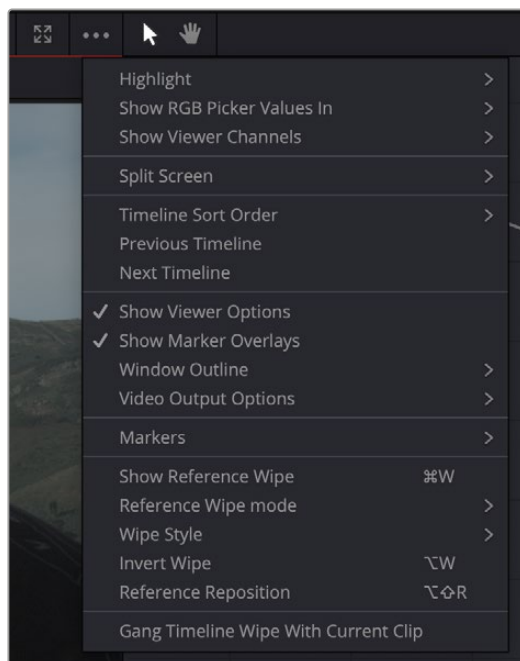
The Viewer title bar has controls and indicators making it easy to control and keep track of what it is you're looking at.



The Color page title bar with its controls

The Viewer title bar has the following controls:

	Zoom and Fit menu: Lets you zoom to a specific percentage, or choose Fit to fit the image to the total available area of the Viewer.
	Viewer Modes: Lets you switch the Viewer type to Image Wipe, Split Screen, or Highlight modes.
	Playback frame rate indicator: A dot shows green if playback performance matches the frame rate of the project, or red if playback performance drops below real time. To the right, the current frame rate is displayed.
	Timeline name and selection drop-down: The name of the currently open timeline is displayed. A drop-down to the right lets you open any other timeline in the current project to take its place.
	Timecode viewer and drop-down: A second timecode viewer lets you choose an alternate timecode/frame count/KeyCode value to display simultaneously to the timecode viewer next to the transport controls below. A drop-down lets you pick whether to display source (clip) or record (timeline) timecode.
	Proxy Handling: Lets you choose to use either proxies or original media for playback.
	Bypass Color and Fusion and drop-down: Lets you disable grades and/or Fusion effects.
	Expand Viewer drop-down: Expands the Viewer to take up the full area of your workstation's display above the palettes.
	Option menu: Has options that affect the Viewer's functionality.



The Color Viewer's option menu, showing all the Viewer-related commands

Highlight: Has options for activating the Highlight, Black and White, and Difference views.

Show RGB Picker Values In: Select if the RGB picker shows values in 8-bit or 10-bit scale.

Show Viewer Channels: Has options to select all or individual R, G, or B channels in the Viewer.

Split Screen: Has options for turning on and off and what to display in the Split Screen view.

Timeline Sort Order: Has options for how to order the timelines in the drop-down menu at the top of the Viewer. Options are: Alphabetical, Creation Date, or Recently Used (shows the last 10 actively used timelines).

Previous Timeline: Opens the previous timeline in the Timeline drop-down menu list.

Next Timeline: Opens the next timeline in the Timeline drop-down menu list.

Show Viewer Options: Show or hide the second row of icons at the top of the Viewer, depending on the mode you're in.

Show Marker Overlays: Markers that intercept the playhead when playback is paused appear superimposed in the Viewer.

Window Outline: Has options on how to display the window outline for a Power Window. On, Off, or only on the UI viewer and not on your video output.

Video Output Options: Has options to choose what to display on your video output signal, Flags, Markers, Keyer Overlays, and to Gang the Zoom level with the Viewer.

Markers: Displays any markers that have been set on the timeline. Select one in the list to go to that marker.

Show Reference Wipe: Turns on or off the reference wipe.

Reference Wipe Mode: Has options to change what sources are used in the reference wipe.

Wipe Style: Has options to change what type of reference wipe is used.

Invert Wipe: Inverts the current reference wipe.

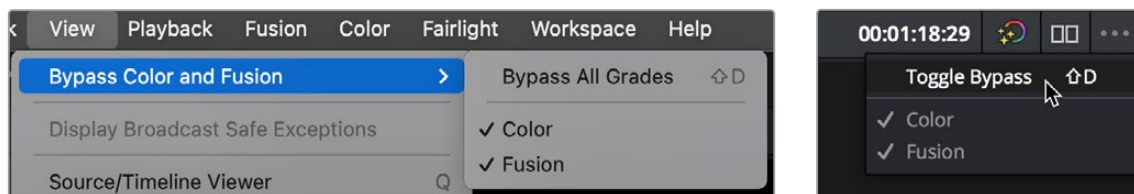
Reference Reposition: Opens the Reference Sizing controls that allow you to resize and move the reference clip.

Gang Timeline Wipe With Current Clip: Lets you maintain the offset between the current clip and a timeline clip you're wiping against when you move the current clip selection to other clips.

Turning Grades and/or Fusion Effects Off

The Bypass Color Grades and Fusion Effects button/drop-down commands in the Viewer's title bar are also available via View > Bypass Color and Fusion menu commands. Turning off Fusion effects in the Color page is an easy way to improve playback performance on low power systems when you just need to make a quick set of grading adjustments. Toggling grades off and on is also a convenient way to quickly get a before-and-after look at a shot where the "before" goes all the way back to the source.

If you choose Toggle Bypass or click the Viewer control, you'll turn off whatever is checked in the optional menu, which lets you choose whether or not you want to bypass both Color and Fusion, or just one or the other.



(Left) Menu commands for bypassing Color and Fusion, (Right) Edit page Timeline Viewer controls

TIP: If you're giving your client a before and after look at work you're doing on a grade, a more effective technique is to select the specific nodes (one or more) that you want to toggle on and off, and press Command-D (Enable/Disable Selected Nodes.)

Viewing Isolated Channels

Use the Show Viewer Channels submenu in the Viewer's option menu to switch the Color page Viewer among RGB, R, G, or B channels. This can be useful when evaluating a single channel of an image for noise or other artifacts, or for doing color matching by comparing and adjusting the individual red, green, and blue channels of two different clips.

The Viewer Toolbar

The Color page also exposes a toolbar at the top of the image (underneath the title bar) that makes it easy to enable and disable image wipe, split screen, and highlight viewing by clicking one of these buttons, which then exposes additional controls relating to each of these modes of operation (described elsewhere in this chapter). Those options can be shown or hidden by clicking the Viewer option menu and unchecking Show Viewer Options.

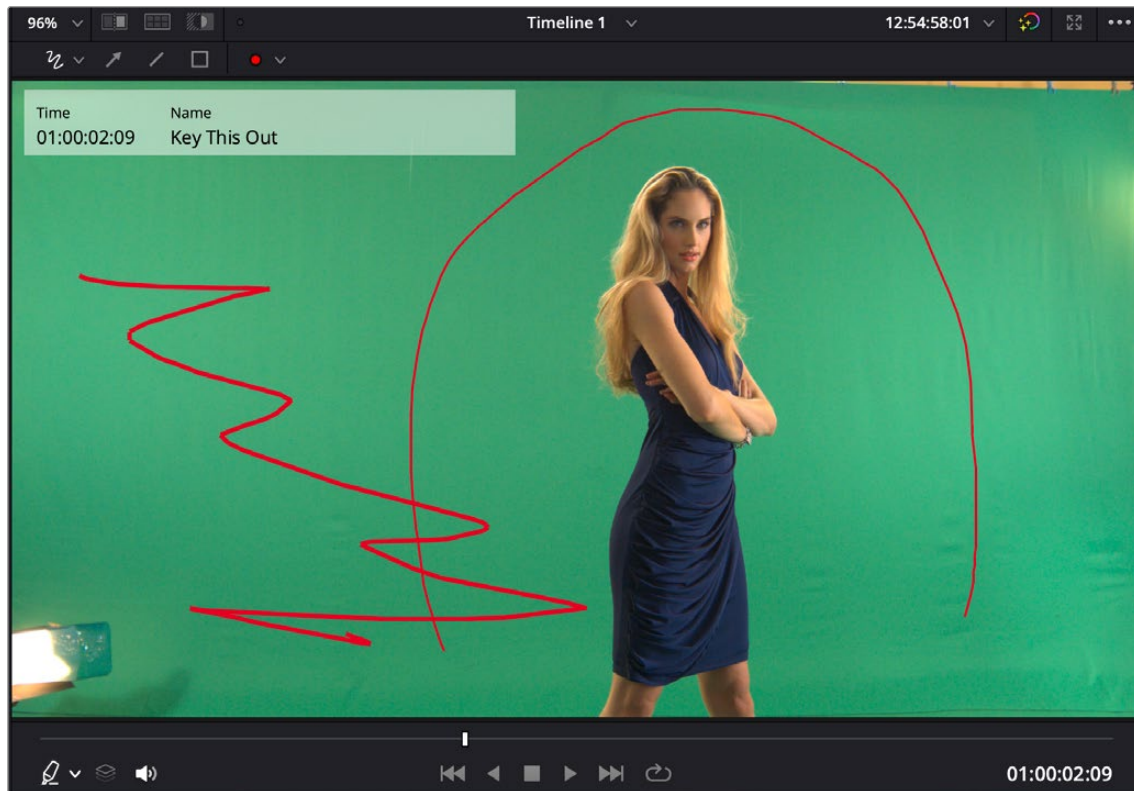


The Color page Viewer toolbar showing the Wipe, Split Screen, and Highlight controls. The Highlight options are seen in the bar below.

In the process, these buttons provide an easy reference for when a comparison mode is enabled. In each of these Viewer modes, the appropriate controls for customizing that view appear to the right of the Viewer toolbar.

Showing Marker Overlays and Annotations

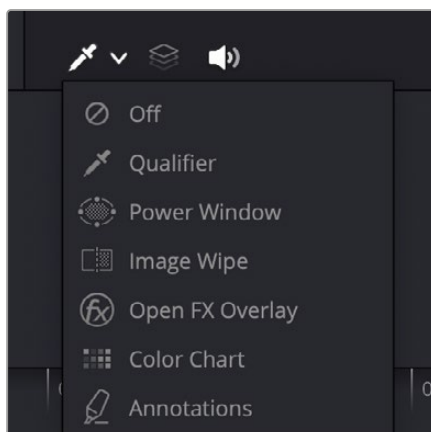
The Color page Viewer supports viewing Marker Overlays and Annotations for Timeline and Clip markers. From the Viewer's 3-dot option menu select Show Marker Overlays, or from the Viewer overlay controls drop-down (bottom left) select the annotation mode.



When Show Marker Overlays is checked, the Color Viewer will display any Marker Overlays (upper left) and annotations.

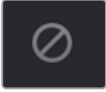
The Onscreen Control (OSC) Menu

The Onscreen Control drop-down menu lets you choose which onscreen controls you want to display and adjust in the Viewer. Some palettes automatically enable corresponding onscreen controls when you open them. For example, opening the Window palette displays the Power Window onscreen controls, and opening the Qualifier palette displays the Qualifier onscreen controls.



The onscreen control menu for the Viewer

There are a variety of choices, each corresponding to different adjustments you can make:

	Off: All onscreen controls are hidden from view, giving you an unimpeded display of the image in the Viewer.
	Qualifier: Turns on the Color Sample cursor. For more about using Color Sample controls, see <i>Chapter 135, “Secondary Qualifiers.”</i>
	Power Window: Turns the Power Windows onscreen control on and off. For more about adjusting windows in the Viewer, see <i>Chapter 136, “Secondary Windows.”</i>
	Image Wipe: Toggles mouse control of dragging the split screen directly on the Viewer on and off. For more about working with split screens, see <i>“Comparing Clips in the Viewer”</i> later in this chapter.
	Open FX Overlay: Shows and hides whatever onscreen controls are exposed by an Open FX plugin in the currently selected node.
	Color Chart: Shows and hides the color chart overlay that lets you identify a color chart in the picture that you want to use with the Color Match palette to create an automatic grade.
	Annotations: Shows the annotation tools in the toolbar. Drawing annotations on a frame without a marker creates a new Timeline marker. For more about using Annotations, see <i>Chapter 41, “Marking and Finding Clips in the Timeline.”</i>

TIP: As you work, you may find you want to temporarily hide or show the onscreen controls so you can get an uncluttered look at the image you’re adjusting. You can toggle any set of onscreen controls off and on without selecting Off in the menu by pressing Shift-` (tilde).

Toggling Viewer Overlays On and Off

Pressing Shift-` (Tilde) turns the current Viewer overlay on and off. The View > Viewer Overlay submenu contains commands for showing, hiding, and switching different overlays in the Color page Viewer. While most of the options in this menu are unassigned by default, they can be assigned to keys using the Keyboard Customization window.

Onscreen Controls and External Displays

DaVinci Resolve has been designed for use with calibrated external displays connected to video output interfaces, and for most colorists working on broadcast or theatrical programs, this is the recommended way to work for color critical evaluation.

Because of this, many of the onscreen controls associated with tasks such as color sampling, window adjustment, and key manipulation are mirrored to your video output, making it possible to hide the Viewer on your computer’s monitor and work with only an external display.

To choose whether onscreen controls are mirrored to video out or disabled:

Choose an option from the Window Outline submenu in the Viewer's option menu. There are three options:

- **Off:** Hides the window outline on both the external display and the Viewer.
- **On:** The default, shows the window outline on both the external display and the Viewer.
- **Only UI:** Hides the window outline on your external display, but leaves it in the Viewer.

To show clip flags on video output:

Turn on the "Show clip flags on video output" checkbox option in the Color page Viewer's option menu. This enables drawing one or more small colored flag overlays in the bottom-left corner of video output. This way, you can always see which clips are flagged on your grading hero display as you work, if that helps you to keep track of whatever you're using flags to stay on top of. This is off by default.

Limitations When Grading With the Viewer on a Computer Display

Most computer displays do not operate at the color critical tolerances or specifications required for quality control for streaming, broadcast, or theatrical delivery. An additional issue, however, is that depending on your combination of workstation and computer display, the Viewer does not necessarily display each clip's image data as it is displayed by the calibration that your operating system applies to your computer display, depending on which OS you're running DaVinci Resolve on. This makes your computer display potentially unsuitable for monitoring projects destined for the sRGB standard of the web in its default state. For example, if you grade a project using the Color page Viewer on your computer display, the resulting clip may not look the same in the QuickTime player, or in other post-production applications.

You can address this in one of two ways:

- If you're using DaVinci Resolve on macOS, you can turn on "Use Mac Display Color Profile for viewers" in the Hardware Configuration panel of the System Settings. This lets DaVinci Resolve use whichever of the color profiles you choose in the Color tab of the Displays panel in the macOS System Preferences, thereby taking advantage of ColorSync on macOS to let DaVinci Resolve display color the way your computer monitor does. This now works for all color profiles that ship with macOS, as well as color profiles that have been generated by calibration software, such as that available from X-rite, Datacolor, or other applications. On supported computers, there's also an option to "Use 10-bit precision in viewers," if available, that you can turn on. With this option enabled, rendered output displayed in QuickTime Player will match what is seen in the DaVinci Resolve Viewer.
- Alternately, you can apply a dedicated Color Viewer LUT for calibration, using the 1D/3D Color Viewer Lookup Table drop-down menu that's found in the Color Management panel of the Project Settings. This lets you analyze your computer display for calibration in the same way you would calibrate an external display, using a probe and color management software, and apply the resulting calibration LUT in DaVinci Resolve. Keep in mind that monitor calibration can only make a high quality display standards compliant; it cannot make up for a display gamut that's too small. For more information, refer to "Lookup Tables" section see *Chapter 4, "System and User Preferences."*

Strictly speaking, if you're doing professional work, you should restrict your grading using a calibrated, 10- or 12-bit class A external broadcast display of some type, connected via a Blackmagic Design video interface. Assuming everything is running properly, an image that is output to video from DaVinci Resolve should match an image output to video from any other post-production application you're using, and this should be your basis for comparison when examining the output of two different applications.

Zooming into the Viewer

When using the Viewer to sample tricky colors or draw a detailed PowerCurve, it can be advantageous to zoom into the image.

To zoom into or out of the Viewer, do one of the following:

- Move the pointer to within the Viewer, and then roll the scroll wheel to zoom in or out of the image.
- Press Command-Equal to zoom in, or Command-Minus to zoom out.

To pan around the Viewer:

- Move the pointer to within the Viewer, then middle-click and drag to pan around the image.

To reset the size of the Viewer image:

- Choose 100% (Actual Size) (Option-Shift-Z) from the sizing drop-down menu in the upper left of the Viewer.
- Choose View > Zoom > Zoom to Fit (Shift-Z).

To enable or disable image zoom being sent to video out:

- Choose “Gang viewer zoom with video output” from the option menu of the Viewer to zoom the image shown on video out identically to the zoom level of the Viewer, enabling you to evaluate a zoomed-in portion of the image on your suite’s hero display.

Using the Jog Bar and Transport Controls

One of the principal uses of the Viewer is to control playback. The jog bar, directly underneath the image in the Viewer, contains a playhead that you can drag to the left and right to navigate quickly through the currently selected clip as fast as you can move the pointer. The playhead in the jog bar is locked to the playheads found in the Timeline and Keyframe Editor. Moving one playhead moves all three.

How much of the Timeline the jog bar navigates depends on whether the Node Editor is set to Clip or Timeline mode. In Clip mode, the jog bar width equals the duration of the currently selected clip. In Timeline mode, the jog bar width equals the total duration of the entire Timeline.

A row of transport controls below the jog bar provides more specific control over timeline playback.

- **Previous clip:** Moves the playhead to the first frame of the previous clip.
- **Reverse:** Initiates 100% playback in reverse.
- **Stop:** Stops playback.
- **Play:** Initiates 100% playback.
- **Next clip:** Moves the playhead to the first frame of the next clip.
- **Loop:** Lets you restrict playback to the current clip, looping to the first frame if you’re playing forward to the end of a clip, or looping to the last frame if you’re playing in reverse to the beginning of a clip.

Two other buttons let you control audio playback and clip display in the Viewer:

- **Unmix:** Turning on Unmix disables all transitions, composited superimpositions, and effects that mix two or more clips together. Unmix allows you to judge the appearance of a clip without distraction whenever you need to make an adjustment based on how the clip looks on its own, or whenever you need to make changes based on frames that would otherwise be hidden underneath a transition such as a dissolve or fade from black. When you're ready to see how your grades work in context with transitions and composites again, turn Unmix off.
- **Mute:** Audio playback can be turned on or off by clicking on the speaker icon, or adjust the level by right-clicking on the speaker icon and dragging the slider.

Navigating Using the Arrow Keys

You can use the Arrow Keys of the keyboard to navigate clips and the Timeline in different ways.

- **Up/Down Arrow:** Moves the playhead to the first frame of the previous or next clip.
- **Left/Right Arrow:** Moves the playhead back or forward one frame at a time.
- **Shift-Left/Right Arrow:** Moves the playhead back or forward one second at a time.

Controlling Playback Using the Spacebar and JKL Keys

You can also use the spacebar to start and stop playback, or the JKL keyboard shortcut convention for controlling playback, where J plays in reverse, K stops playback, and L plays forward. There are many additional uses of these keyboard shortcuts.

For more information see *Chapter 35, "Preparing Clips for Editing and Viewer Playback."* in "Using JKL to Control Playback"

Fast Review on the Color Page

The Fast Review playback command (Playback > Fast Review) is now available on the Color page. Intended to help you watch through long sequences of clips quickly, clicking this option begins accelerated playback through the Timeline, where the speed of playback is relative to the length of each clip you're playing through. Long clips play faster, whereas shorter clips play closer to real time.

Navigating Using Timecode

You can also use absolute or relative timecode entry to move the playhead in the Color page Viewer. When entering timecode, type each pair of hour, minute, second, and frame values consecutively, with a period representing a pair of zeros for fast entry. The last pair of timecode values (or period) you enter is always assumed to be the frame number, with any untyped values assumed to be zero. It's not necessary to enter colons or semicolons.

For more information, see *Chapter 35, "Preparing Clips for Editing and Viewer Playback."* in "Moving the Playhead Using Timecode."

Viewer and Transport Timecode Displays

The Viewer has two timecode displays, each of which defaults to a different timecode setting. A timecode display at the top, the Header timecode display, shows the Source timecode by default. The timecode display at the bottom, the Transport timecode display, shows the Record timecode by default. The bottom timecode display can be changed to show one of four different options, which are similar to those found in the Data Burn-In palette.

To change a timecode display to a different setting:

The top timecode display is a drop-down menu that can be changed to whatever timecode, frame number, or KeyCode you want to display. The bottom timecode display can also be changed by right-clicking on it and choosing the type of value to show from the contextual menu that appears. There are the following options:

- **Timeline Timecode:** The timecode corresponding to the playhead's position in the overall Timeline.
- **Source Timecode:** The timecode corresponding to the playhead's position relative to the currently selected clip.
- **Timeline Frame:** The frame count corresponding to the playhead's position in the overall Timeline.
- **Source Frame:** The frame count corresponding to the playhead's position relative to the currently selected clip.
- **KeyCode:** The KeyCode number corresponding to the media's KeyCode track, if there is one.
- **Show Timecode at 30 FPS:** Displays 24 fps timecode, via 3:2 pulldown, as 29.97 fps timecode. Has no effect on video playback.
- **Copy and Paste Timecode:** Two commands make it easy to copy and paste timecode values.

In the same way, the top timecode display can alternately be changed to show KeyCode, if it's available, within a DPX media file's header.

Enhanced, Full, and Cinema Viewing

You can expand the Viewer into Enhanced Viewing mode by choosing Workspace > Viewer Mode > Enhanced Viewer Mode (Option-F).

Enhanced Viewing Mode

In Enhanced Viewing mode, the Viewer works exactly as it does at its regular size, but it expands to fill up the entire area of the screen above the palettes and Keyframe Editor. To exit Enhanced Viewing mode, press (Option-F) again.

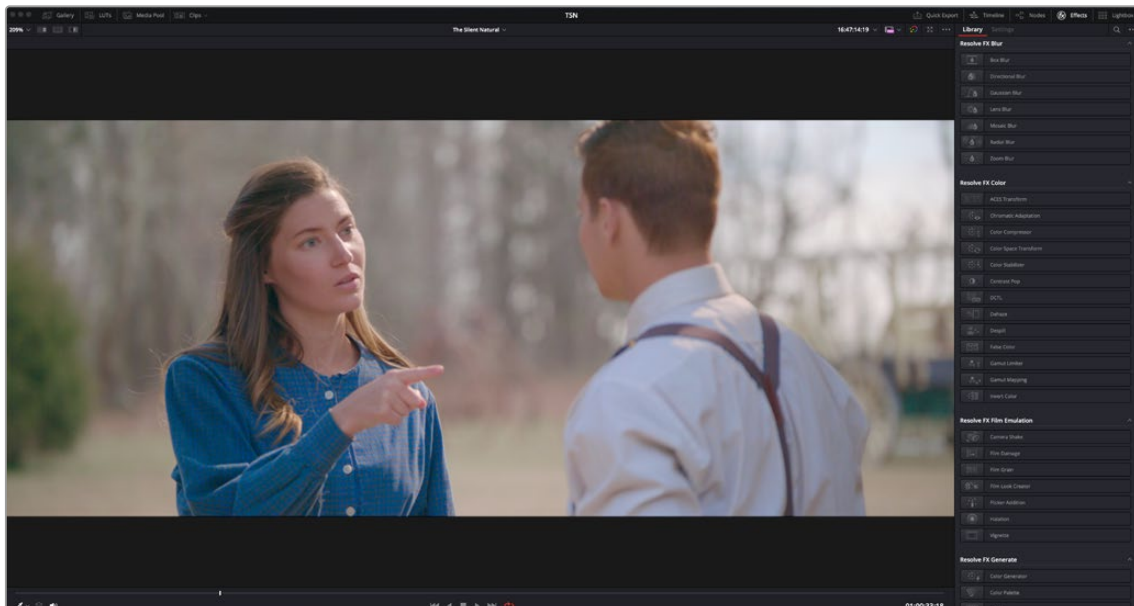
This can be useful if you need a closer view of the image for purposes of making detailed corrections, examining noise patterns up close, making a tricky color selection, or drawing a complicated PowerCurve.



Color page in Enhanced Viewer mode

Full Viewing Mode

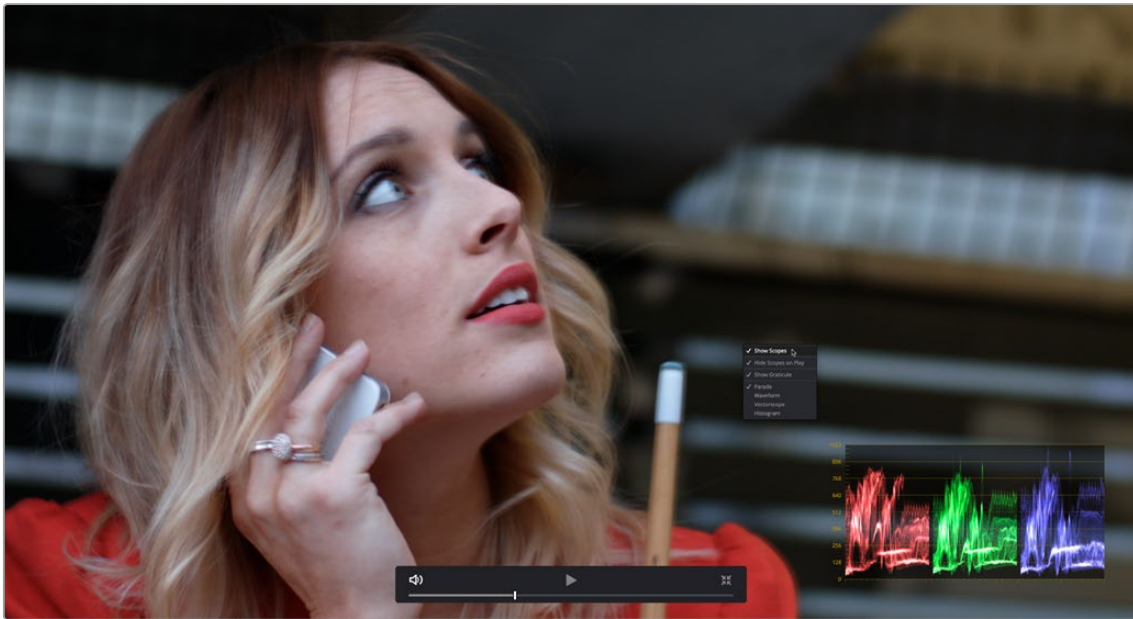
In Full Viewing mode (available by choosing Workspace > Viewer Mode > Full Page Viewer, pressing Shift-F, or Option-clicking the Enhanced Viewing mode button), the Viewer takes up even more room by hiding the palette controls, but leaves room for the transport controls, the Onscreen Control drop-down menu, the timecode display, and the page buttons along the bottom of the DaVinci Resolve UI. This mode is useful when you need an even closer look at the image, but you still want access to a minimal set of onscreen controls.



Color page in Full Viewer mode

Cinema Viewing Mode

Another option, Cinema mode, allows for full-screen viewing. Cinema mode is available by choosing Workspace > Viewer Mode > Cinema Mode Viewer (P). In Cinema mode, the menu bar is hidden, and the image is presented full screen without any of the ordinary onscreen controls. Moving the pointer over the screen reveals a set of hidden onscreen controls that include a play button, jog bar, mute button, and exit button (which lets you turn off Cinema mode). You also have the option of superimposing scopes over the image, which can be valuable in on-set situations.



Color page in Cinema mode

Cinema mode is useful for doing detailed reviews of media in on-set and digital dailies workflows when working remotely without a secondary video display.

TIP: If you'd like to superimpose timecode over the image in Cinema mode for reference, you can use the controls within the Data Burn-In window (Workspace > Data Burn-In) to set up whatever information you'd like to display during playback.

Safe Area Overlays in the Viewer

You can show or hide a series of safe area overlays in the Viewer. All safe area overlay options are found in the View > Safe Area menu. Each safe area overlay option can be individually enabled or disabled from this menu.



Viewer displaying extents, center, safe, action and title area overlays

On/Off: Turns all currently selected safe area marker options on or off at once.

Extents: An outline showing the exact outer edge of the frame. Especially useful when the safe markers are set to an aspect ratio other than that currently used by the Viewer.

Action: An outline showing the outer 90% action safe area of the frame.

Title: An outline showing the outer 80% title safe area of the frame.

Center: A crosshairs showing the center of the frame.

Aspect: Enables use of the View > Select Aspect Ratio submenu to change the aspect ratio of the safe area markers. You can choose among the following aspect ratios: 1.33 (a.k.a. 4:3), 1.66, 1.77 (a.k.a. 16:9), 1.85, and 2.35.

Use Gray Background

Checking the “Use gray background in viewers” box in the User Preference’s UI Settings sets the empty area of the Viewer (if there is any) to a lighter gray, making it easier to see which parts of the Viewer are black because of blanking, and which parts are simply empty because of the way the image is zoomed or panned.

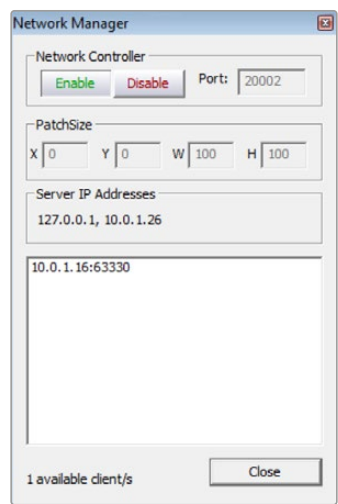
Monitor Calibration

If you use Light Illusion's LightSpace CMS or SpectraCal's Calman for display calibration, you can now use DaVinci Resolve as a sync-able pattern generator. This means you can use DaVinci Resolve to output color patches, synchronized by LightSpace, to your display through whichever video interface is connected to your computer. These synchronized color patches will be analyzed by a monitor probe that's also controlled by LightSpace, which stores the probe data and compares it to the original color values of the output color values in order to characterize that display.

To use this feature, you must first have a licensed copy of LightSpace CMS, which is a Windows application. Synchronization depends on a wired or wireless LAN being available to connect the LightSpace application with DaVinci Resolve.

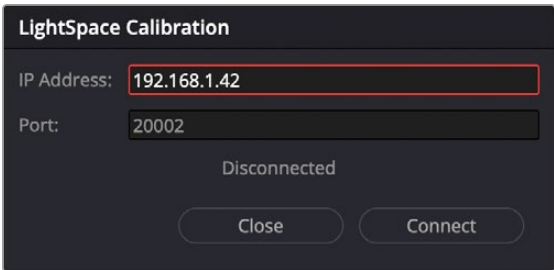
To synchronize LightSpace CMS to DaVinci Resolve as a pattern generator client:

- 1 Open LightSpace on the Windows computer that's running it.
- 2 When LightSpace is open, click the Network Manager button. A window appears displaying the two network IP addresses used by LightSpace. Note these, and click the Enable button.



The Network Manager dialog in LightSpace CMS

- 3 In DaVinci Resolve, choose Workspace > Monitor Calibration > LightSpace.
- 4 When the LightSpace dialog opens, enter the second of the two network IP addresses LightSpace lists into the Remote Machine field, and make sure the Port number matches. Then click the Connect button.



The Calibration dialog in DaVinci Resolve, connected to LightSpace CMS via WiFi

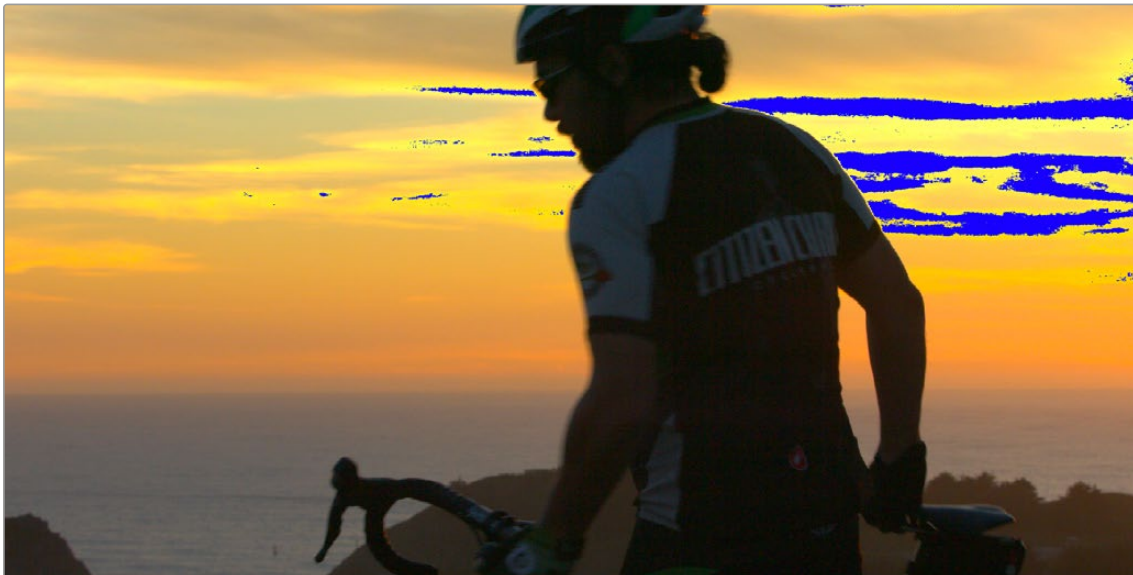
If everything is working correctly, the LightSpace dialog in DaVinci Resolve should show the word “Connected,” and the Network Manager dialog in LightSpace should show that there is 1 available client/s.

You can now close the Network Manager dialog in LightSpace and follow the procedures outlined in the LightSpace CMS documentation for characterizing your display and building and exporting a display LUT (in the .cube format) that you can use as a display LUT in DaVinci Resolve. Alternately, you can export a display LUT from LightSpace that can be loaded onto an outboard video processing device.

- 5 When you’re finished, click Disconnect in the LightSpace dialog, and then click Cancel to close the window.

Viewing Broadcast Safe Exceptions

Choosing View > Display Broadcast Safe Exceptions sets the Color page Viewer to show a false-color overlay that indicates in blue the regions of the picture that violate the currently selected broadcast safe level in the Color Management panel of the Project Settings.



Areas of the image that violate broadcast safe are highlighted in blue

More information about broadcast safe limiting in the Color Management panel of the Project Settings see *Chapter 127, “Automated Grading Commands and Imported Grades.”*

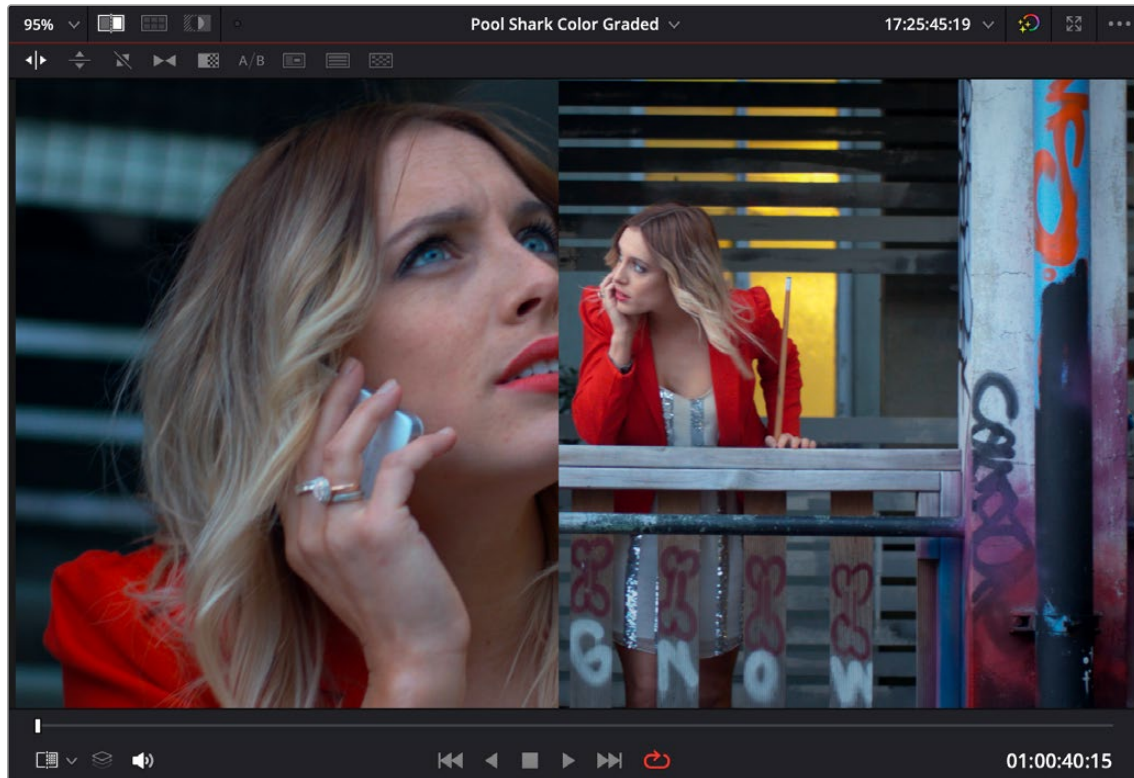
Comparing Clips in the Viewer

The ability to compare different clips to one another is an important part of the color correction process. DaVinci Resolve provides three different ways of doing so. You can use the Gallery to display two clips for split screen comparison. You can also use different reference modes to see a timeline clip or reference movie directly as part of a split screen comparison. Finally, you can use the Split Screen controls to display multi-frame arrangements in the Viewer.

Saving and Wiping Stills in the Gallery and Timeline

The Gallery on the Color page provides fast access to stills that you've saved from various clips in the Timeline. While the dedicated Gallery page provides a more comprehensive interface for browsing presaved “looks,” as well as for importing stills from other projects, you can save, organize, and browse stills directly within the Gallery of the Color page.

Stills are saved in the DPX file format. Once you've saved one or more stills, you can set up split screen wipes in the Viewer, which will be mirrored to your external display.



Stills from the Gallery can be compared to the current shot, making it easier to match grades.

This section provides an abbreviated summary of still store and split screen functionality to get you started quickly.

To save a still, do one of the following:

- Choose Color > Stills > Grab Still (Option-Command-G).
- Right-click on the Viewer and choose Grab Still.

To wipe a still, do one of the following:

- Select a still in the Gallery, and click the Image Wipe button on the top Viewer toolbar.
- Choose Color > Stills > Play Still (Command-W), or right-click in the Viewer and choose Toggle Wipe.
- Double-click a still in the Gallery.

To adjust a wipe in the Viewer, do one of the following:

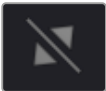
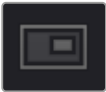
- Drag the pointer within the Viewer to move the wipe.



The different wipe customization modes

To customize a wipe in the Viewer:

Click one of the Wipe mode buttons in the Viewer toolbar. There are the following modes:

	Horizontal: Lets you compare both halves of the wipe to either side of a vertical border. Dragging the pointer moves the wipe border left and right.
	Vertical: Lets you compare both halves of the wipe above and below a horizontal border. Dragging the pointer moves the wipe border up and down.
	Diagonal: Lets you compare both halves of the wipe via an adjustable diagonal border. Dragging the pointer repositions the wipe to the left and right. Hold the Option key down and drag while moving the pointer around in a circle to rotate the border of the wipe to any angle you like.
	Mix: Lets you blend both images together to compare them. Dragging the pointer controls the fade percentage from one image to the other.
	Alpha: Lets you use a qualifier to define transparency in the image of the current clip while comparing it to a Gallery still used as the background. Add a node to the grade and use the qualifier to key a particular color you want to turn transparent (the green of a green screen, for example). Then, connect that node's KEY output to the Alpha Output that appears in the Node Editor. The part of the foreground image isolated by the key becomes transparent, allowing the Gallery still in the background to show through. This can be useful for previewing how the lighting or grade of a foreground VFX plate looks when the image is composited against a particular background image stored in the Gallery.
	Difference (A/B): Lets you view only the pixels that change due to an operation on the node. For example, if you select this highlight and adjust the gain, only the pixels that the gain modified would be visible in the Viewer.
	Box: Lets you view the current clip as a picture-in-picture effect against the still in the background. Dragging the pointer resizes the crop box around the outside of the reference image.
	Venetian Blind: Lets you compare both images being wiped via alternating horizontal strips. Drag the pointer up or down to change the size of the alternating strips. Good for quickly comparing vertical color uniformity, focusing on color and contrast changes that occur from top to bottom.
	Checkerboard: Lets you compare both images being wiped via an alternating checkerboard. Drag the mouse left or right to alternate between images by squeezing each check horizontally. Good for comparing color uniformity across the width and height of two images.

Choose one of the following from the Wipe Style submenu in the Viewer's option menu.

Wipe Style: Cycles among the Horizontal, Vertical, Diagonal, Mix, Alpha, Difference, Box, Venetian Blind, and Checkerboard modes.

Invert Wipe: Reverses each half of the wipe (Option-W).

Hover Scrub in the Gallery

When Live Preview is enabled in the Gallery option menu, the Hover Scrub Preview submenu lets you choose how you want Live Preview to be shown by a thumbnail in the Gallery and in the Viewer when you hover the pointer over a still or LUT in the LUT Browser:

- You can choose to scrub both the thumbnail you're hovering over and the Viewer, letting you preview the current still's grade or a LUT over the duration of the current clip in both the thumbnail and Viewer.
- You can choose to scrub just the thumbnail, leaving the Viewer to show just the grade or LUT over the frame at the position of the playhead.
- You can disable scrubbing altogether, in which case both the thumbnail and the Viewer will only show the grade or LUT over the frame at the position of the playhead.

Copying Grades from Stills in the Gallery

Stills also store the grade from the clip they came from, and can be used to copy grades from one clip to another, or to store grades that you might want to use later.

To copy a grade from a still to a clip, do one of the following:

- Select one or more clips in the Timeline, then right-click a still in the Gallery and choose Apply Grade.
- Select one or more clips in the Timeline, then middle-click a still in the Gallery.

When you copy a still in this way, the saved grade completely overwrites the grade in the target clip, unless you've used the "Preserve number of nodes" option, found in the contextual menu of the Gallery.

For more detailed information on using the Gallery, including options for organizing the Gallery browser, instructions for using Albums, Power Grades, and Memories, as well as other methods of copying grades and performing advanced grade management tasks, see *Chapter 141, "Using the Gallery."* and see *Chapter 142, "Grade Management."*

Different Viewer Reference Modes for Wipes

While it's common to use the Gallery to wipe against reference stills, by changing the reference mode you can also create a wipe against a clip in the Timeline, or against an Offline Reference Clip or Timeline, if one has been assigned to the current Timeline.

To change the Viewer reference mode, do one of the following:

- Choose an option from the Reference Wipe Mode submenu in the Viewer's option menu.
- Choose an option from the Reference Mode icons at the top of the Viewer.

There are three reference modes:

Gallery: The default reference mode. Lets you wipe the current clip against a Gallery still.

Timeline: Lets you wipe the current clip against another clip in the Timeline using the Wipe Timeline Clip command (see below).

Offline: Lets you wipe the current clip against an offline video that's been assigned to the current Timeline.

For more information on assigning Offline References to a timeline for comparison, see *Chapter 55, "Preparing Timelines for Import and Comparison."*

Wiping Between Clips in the Timeline

If you want to wipe the current clip against another clip in the Timeline without saving a still to the Gallery first, there's a special procedure you can follow.

To wipe between two clips using the Timeline reference mode:

- 1 Click the thumbnail of the first clip you want to compare.
- 2 (Optional) A "Gang timeline wipe with current clip" option, available from the Viewer option menu, lets you maintain the offset between the current clip and a timeline clip you're wiping against when you move the current clip selection to other clips. With this option enabled, the offset between the timeline wiped clip and the current clip is maintained when you move the clip selection. You can step the timeline reference clip forward one frame by pressing Command-Option-Right Arrow, and back one frame by pressing Command-Option-Left Arrow. This allows for precise positioning. When this option is disabled, the timeline wiped clip stays where it is regardless of what clip you select.
- 3 Right-click the thumbnail of the second clip you want to compare against, and choose Wipe Timeline Clip from the contextual menu.

Both clips appear split screened against one another.

NOTE: Wipe Timeline Clip only appears in the contextual menu if you right-click on a thumbnail in the Timeline that has not been selected as the current clip.

Change a Timeline Wipe Using the Timelines Album of the Gallery

While you're using the Wipe Timeline Clip to show a wipe of the current clip against any other clip in the Timeline, you can open the Timelines album of the Gallery and click different clips to change which timeline clip you're wiping against (outlined in blue) without changing the current clip (outlined in orange).

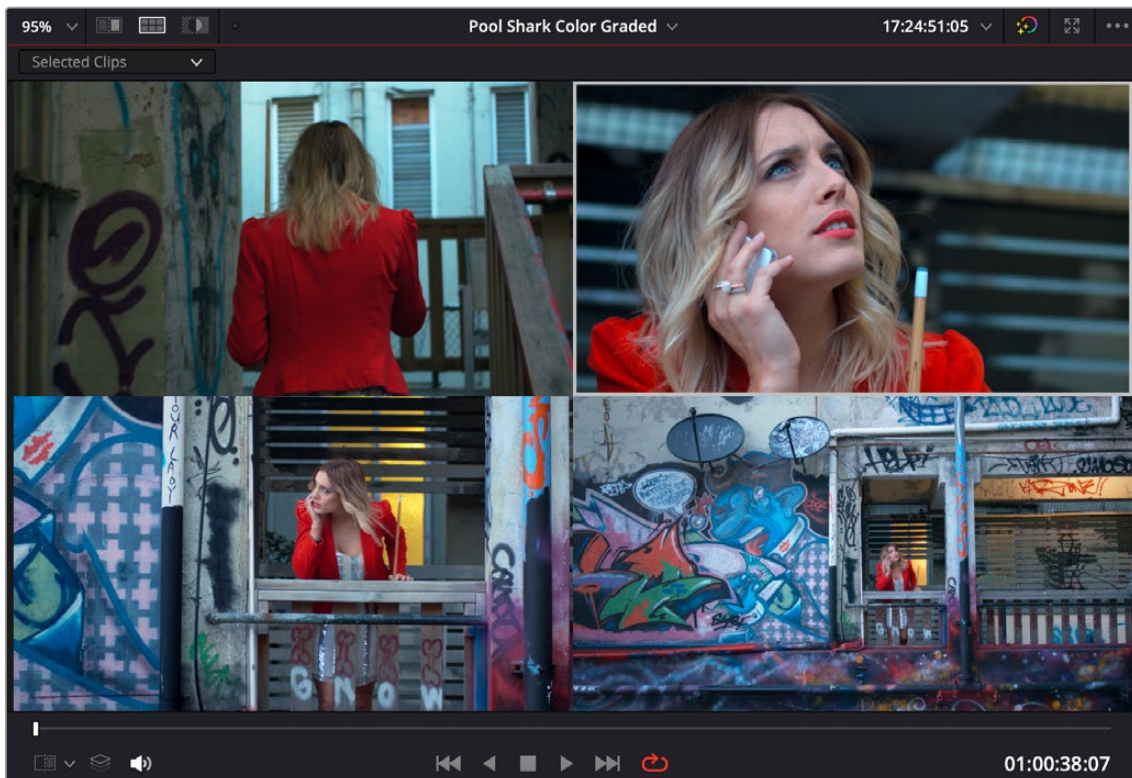
Using Split Screen Modes

A set of split screen modes provides additional ways of comparing multiple images in the Viewer at the same time. These provide side-by-side and grid comparisons of multiple shots and multiple versions, all of which are output to video for monitoring on your external video display. For example, you can use the Split Screen Versions mode to display every single version that's applied to the current clip in a grid, for easy evaluation. The currently selected clip in the Timeline is highlighted in gray, so you can keep track of what you're doing.

To enable a split screen mode, do one of the following:

- Click the Split Screen button on the top Viewer toolbar, and choose a split screen method from the drop-down menu that appears in the Viewer toolbar at the right.
- Choose an option from the Split Screen submenu in the Viewer's option menu.
- Press Option-Command-W to turn Split Screen on and off, using the last selected option.
- Right-click the Viewer and choose an option from the Split Screen submenu in the contextual menu.

NOTE: When any of the split screen modes are selected, you cannot see any other onscreen controls in the Viewer, such as Power Window shapes, until you first disable split screen.



Four shots shown in a grid using the Split Screen > Selected Clips mode

Viewing and Choosing the Current Selection

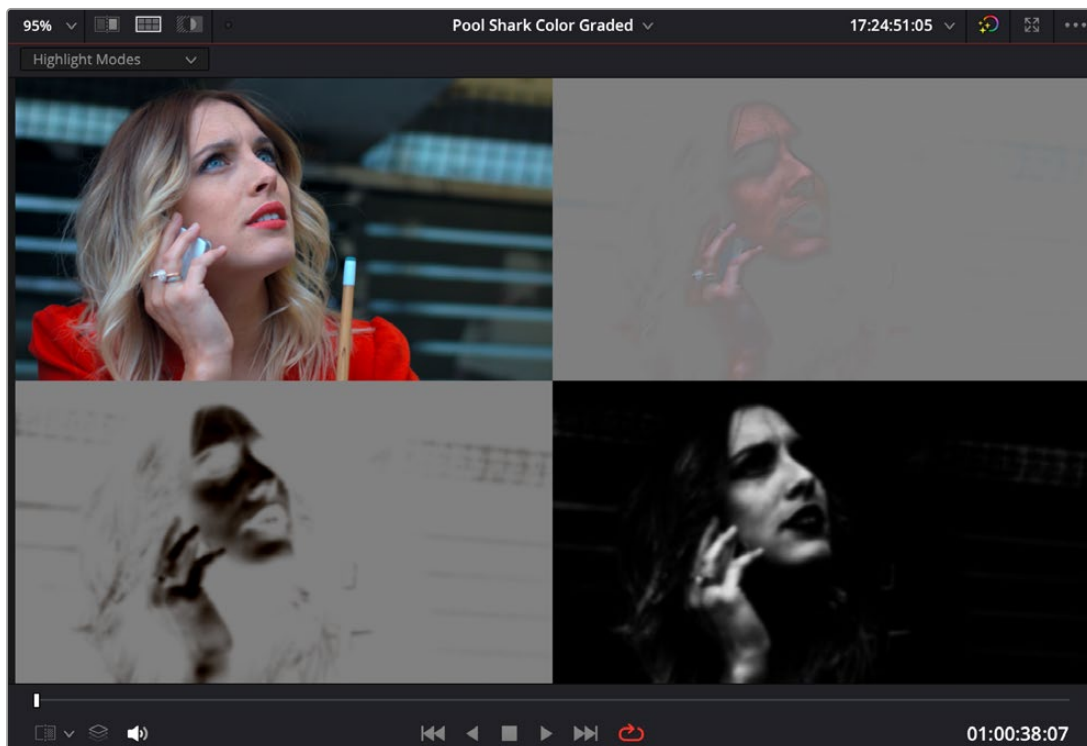
In any of the split screen modes, the current selection is indicated by a gray outline. This outline can be disabled for a more direct comparison by choosing Split Screen > Show Outline in the Viewer's option menu to toggle the outline off and on.

In certain modes, you can also double-click a particular frame of the split screen view to change the current clip, choose the active version, or apply a gallery grade.

Available Split Screen Modes

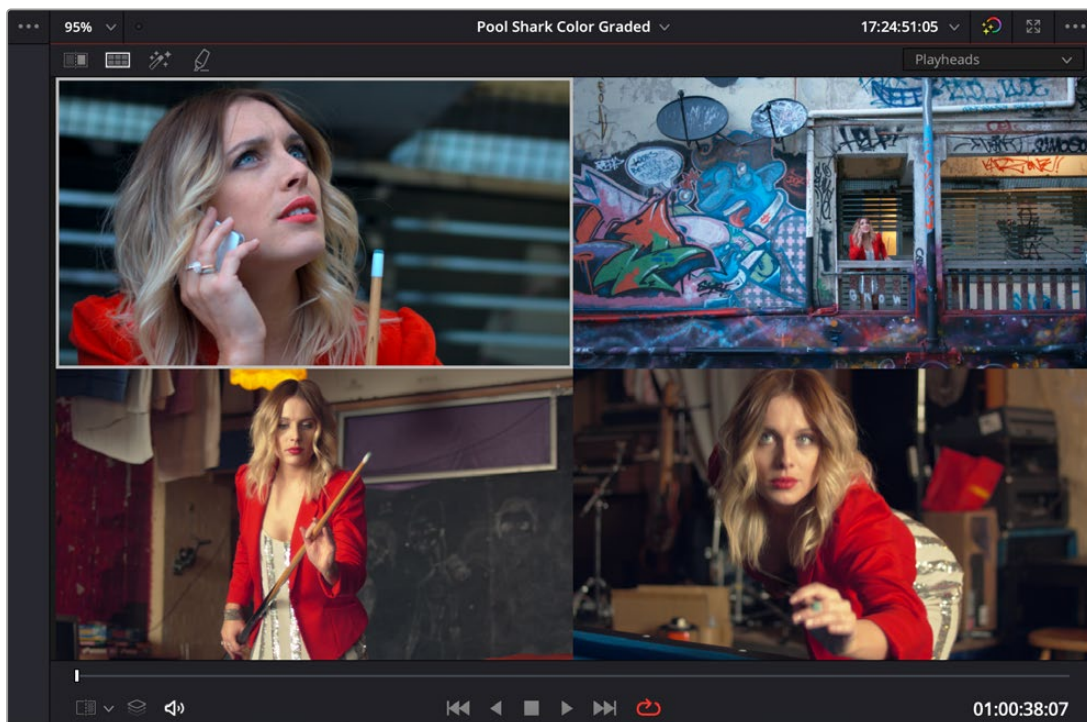
You can choose from among the following split screen modes:

- **Current Group:** Shows up to 16 clips that share the same group as the current clip. Double-clicking a frame does nothing in this mode.
- **Highlight Modes:** Shows a four-up display that lets you simultaneously see (clockwise) the clip's RGB image, the gray matte, the high contrast matte, and a difference matte generated by the node's input compared against its output. Double-clicking a frame does nothing in this mode.



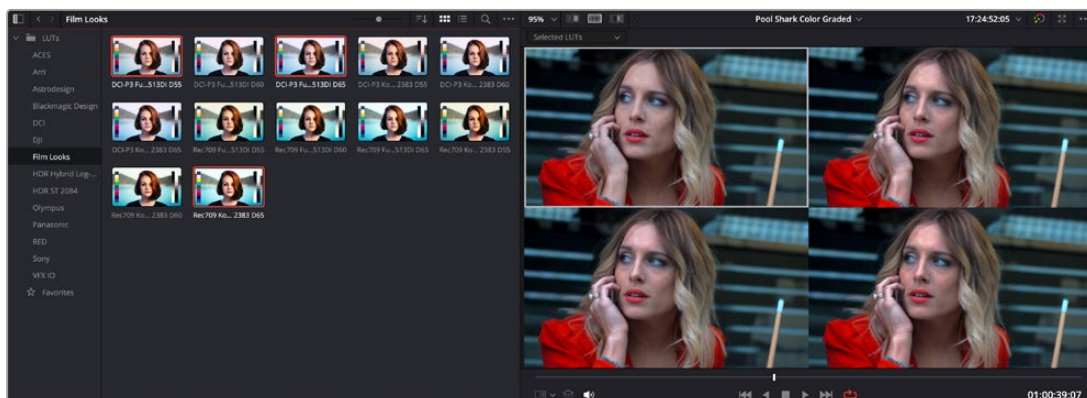
Viewing a key using the Split Screen > Highlight Modes

- **Neighbor Clips:** Shows four clips that immediately surround the current clip. The previous two clips appear as the upper left- and right-hand clips, the current clip appears at the lower left-hand corner, and the next clip appears at the lower right. Double-clicking a frame does nothing in this mode. The “Neighboring Clips in Split Screen” parameter in the Color panel of the User Preferences lets you customize how many neighboring clips appear in this mode.
- **Playheads:** Shows simultaneous comparisons in the Color page Viewer of up to four playheads displayed in the Mini-timeline when you choose multiple playheads from the Color > Active Playhead submenu. For every playhead you expose (up to 4), the frame at each playhead will be displayed in a 2-up, 3-up, or 4-up grid. An outline in the Viewer (if enabled) shows which playhead is currently active. This is useful both for comparing multiple clips, as well as for comparing different frames within the same clip. Switching between the active playheads can be done by clicking on the clips in the split screen Viewer



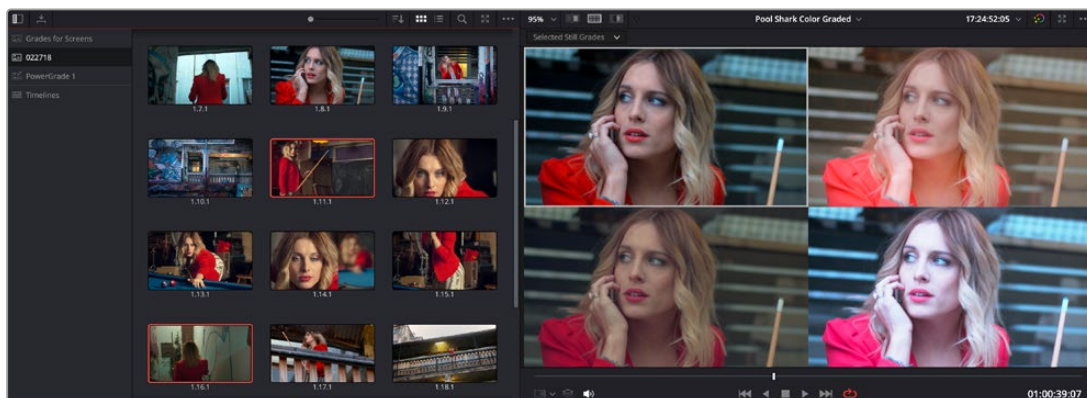
Placing three playheads in the Mini-timeline

- **Selected Album:** When you choose this option, a split screen of the grades of every still in that album are shown in the Viewer as they appear applied to the current clip. Double-clicking a frame applies that gallery grade to the current clip. Up to 16 grades will be shown.
- **Selected Clips:** Shows all currently selected clips in the Timeline, up to 16 at one time. Clicking a clip moves the playhead to the first frame of that clip.
- **Selected LUTs:** Lets you simultaneously display previews of different LUTs effect on the current clip as a split screen in the Viewer by using the LUTs browser to Command-click up to 16 LUTs you want to preview.



The Selected LUTs split screen option lets you preview a bunch of LUT looks at once

- **Selected Still Grades:** When you choose this option, the act of selecting stills in the Gallery of the Color page shows each of the grades associated with those stills as they appear applied to the current clip, in a split screen arrangement. Double-clicking a frame applies that gallery grade to the current clip.



Previewing four gallery grades using the Split Screen > Gallery Grades mode

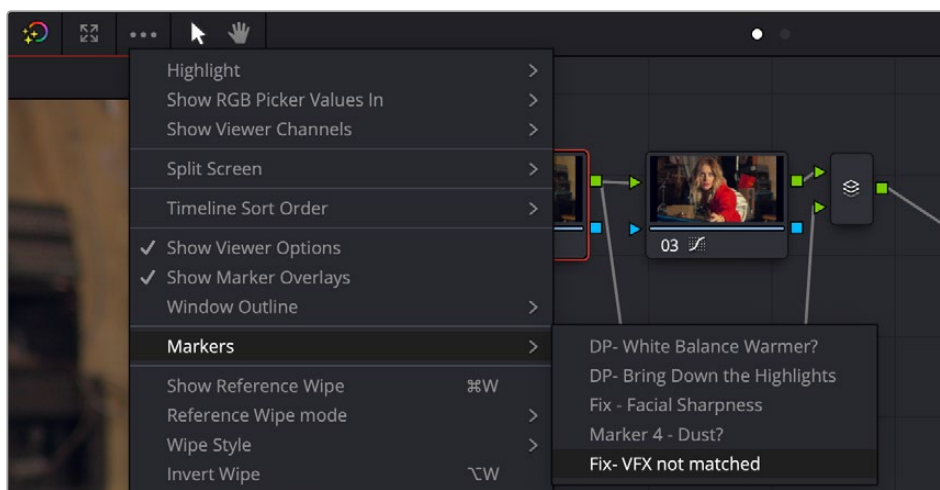
- **Selected Still Images:** Lets you view the current clip against a variety of other stills in the Gallery that you select. This is not a grading preview, you're actually doing a split screen of the selected still images in the Gallery.
- **Version:** Shows up to 16 versions for the current clip, all at once, making it easy to evaluate a series of different versions in relation to one another. If you have more than 16 versions, you deserve some manner of award. Double-clicking a version changes the active version.
- **Versions and Original:** Shows up to the first 15 versions for the current clip, along with the original ungraded clip at the upper left-hand corner. Double-clicking a version changes the active version, but you cannot double-click to select the original.

Marker Overlays and Navigation

If you part the playhead on top of a marker in the Timeline of the Color page, that marker's information appears in a Viewer overlay, just as in the Edit page, making it easier to read notes and see what information is available. These overlays can be easily hidden by clicking the Viewer option menu and turning off "Show marker overlays."

Timeline Marker List Available in Color Page Viewer Option Menu

The option menu of the Color page Viewer has a submenu that lists all Timeline Markers in the currently open timeline. This makes it easy to run down client notes.



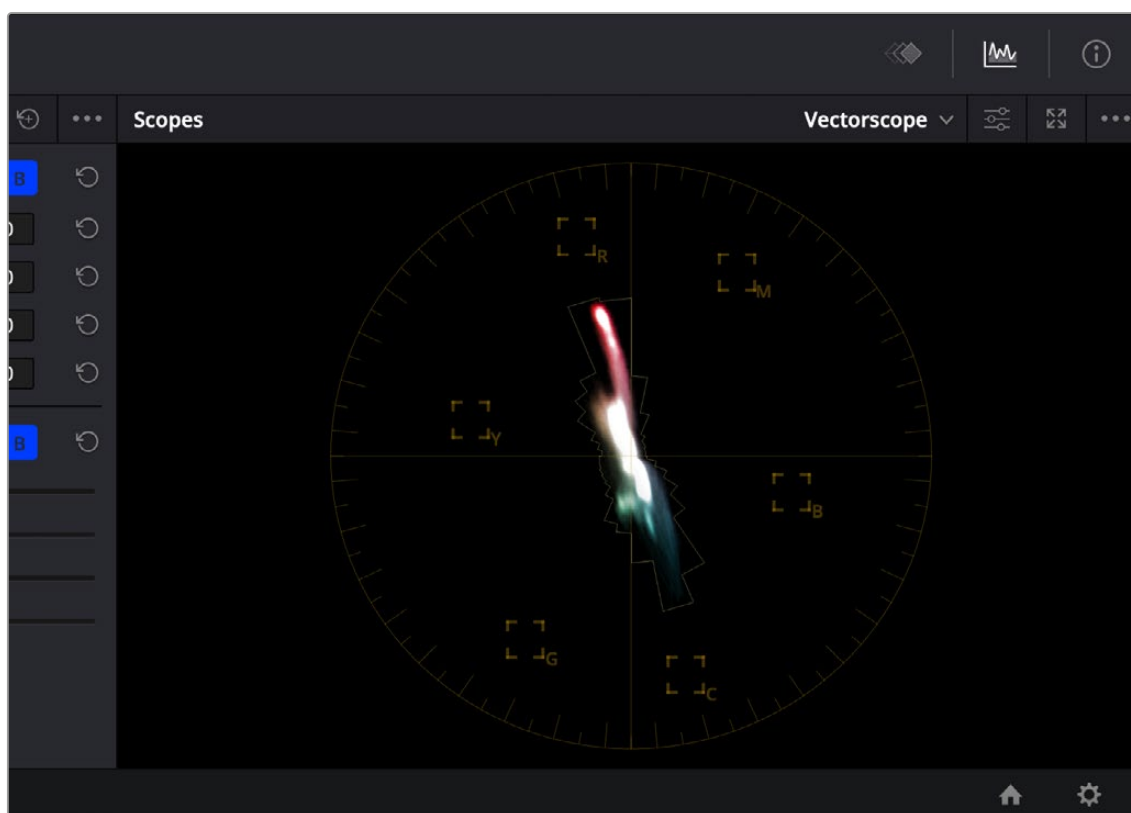
Timeline markers list available for quick access in the Viewer's option menu

Using Video Scopes

While not directly tied to use of the Viewers, video scopes nonetheless work hand in hand to help you evaluate the images you're working on. DaVinci Resolve has a set of five real-time video scopes that you can use to monitor the internal data levels of clips in your project as you work. Each scope provides an unambiguous graphical analysis of different characteristics of the video signal, showing you the relative strength and range of individual color components including luma, chroma, saturation, hue, gamut, white point, and the red, green, and blue channels that, together, comprise the color and contrast of the images in your program.

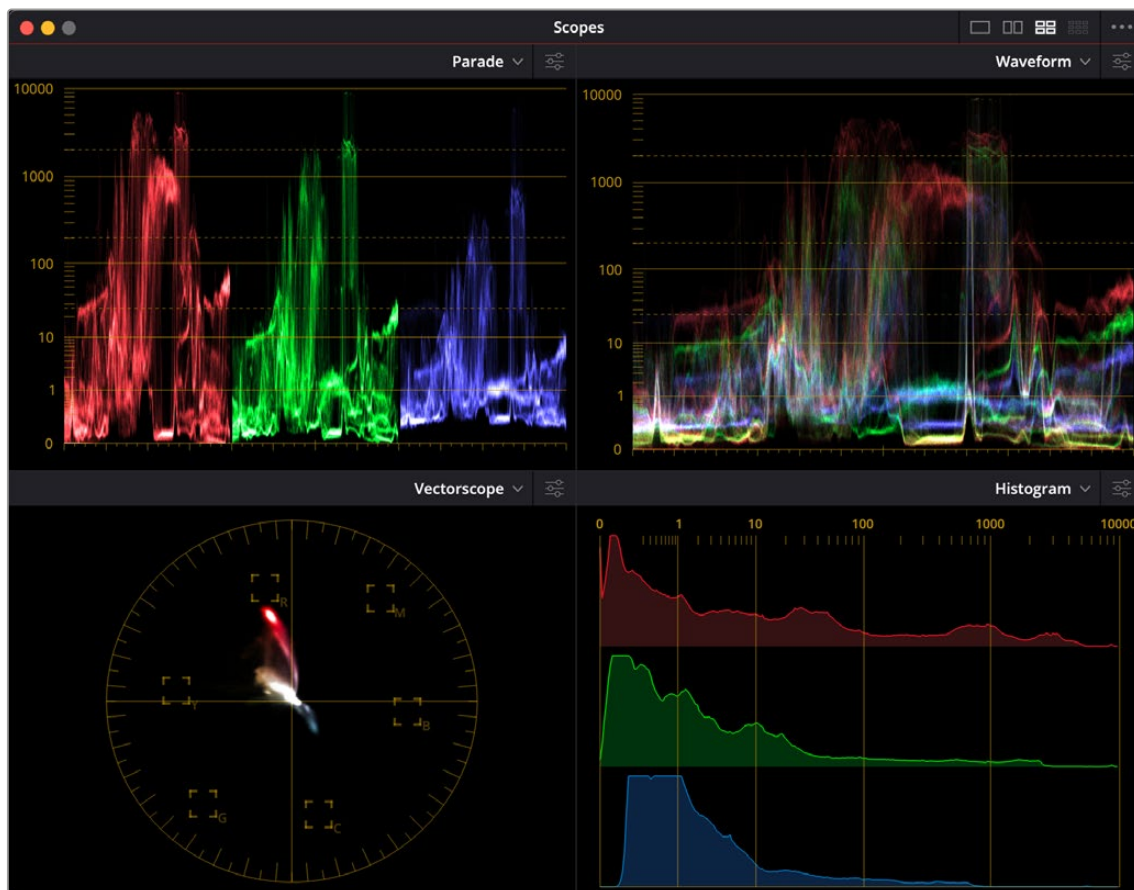
Video Scope Location

By default, you can view any one of these scopes at a time at the bottom right corner of the Color page, by clicking the Video Scope button in the palette button bar.



The video scope, docked next to the other palettes at the bottom of the Color page

Optionally, you can click the Expand button at the top right of the video scope to open the video scopes into a floating window, within which you can display all four video scopes together, or individually, on any monitor connected to your workstation.



Video scopes in a floating window in HDR (ST.2084/HLG) mode

Additionally, the video scopes can appear docked on a second display as part of many of the dual-screen layouts available in DaVinci Resolve. However, if you have three computer displays and you're using dual screen layouts, you still have the option to open the floating Scopes window and place it on your third display, as large as you need.

The video scopes aren't just available in the Color page. They're also available in the Media and Deliver pages for whenever you need to evaluate the video signal more objectively, such as when you're setting up to capture from tape or scan from film, or when you're setting up for output.

To open video scopes from the Media, Color, or Deliver pages, do one of the following:

Choose Workspace > Video Scopes > On/Off (Command-Shift-W) to open video scopes into a floating window.

Choose Workspace > Dual Screen > On to open video scopes as part of a dual screen layout.

Video Scope Measurement Using Scales

Because you're evaluating the internal state of the image data, by default the numeric scales of the WFM and Parade scopes always reflect 10-bit full range data from 0–1023, regardless of the Video/Data Level setting you've selected in the Master Settings panel of the Project Settings. This gives you a window into how the image is being processed by DaVinci Resolve prior to being output via your computer's video I/O interface.

If you're working on an HDR (High Dynamic Range) grade, you can choose HDR (ST.2084/HLG) from the Video Scopes option menu (this feature is only available in Studio version). This replaces the 10-bit scale of the Waveform, Parade, and Histogram video scopes with a scale based on nit values (or cd/m^2) instead.



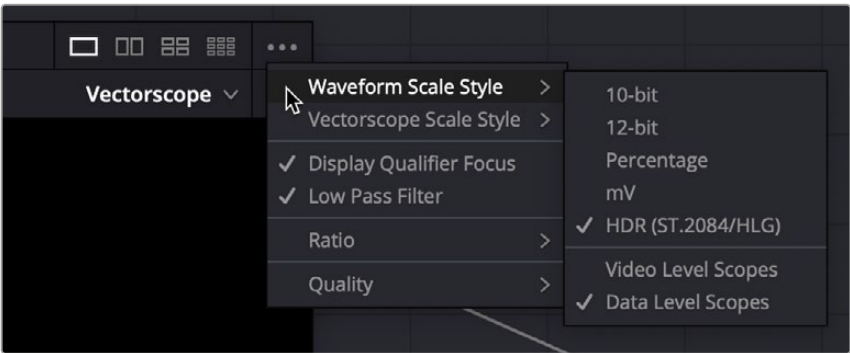
The Parade scope showing the data range scale



The Parade scope showing the HDR "nit" scale

Changing Waveform Scales

A Waveform Scale Style submenu in the Video Scopes option menu (the three dots menu) lets you choose how you want the numeric scale at the left of Waveform and Parade scopes to be represented. There are options for 10-bit display (the default), 12-bit display, Percentage (0 to 100), millivolt (mV), and HDR (ST.2084/HLG). Aside from the added flexibility this gives you, this also means that it's no longer necessary to go to the Preferences to change the scope to have an HDR scale shown in nits (cd/m^2).

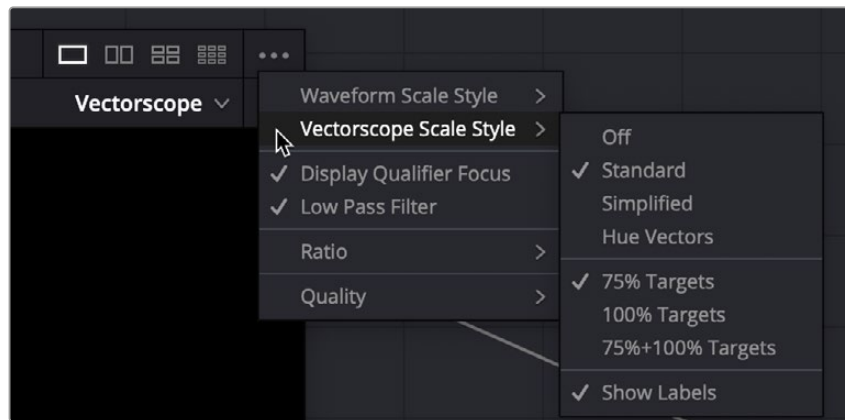


Waveform scale options

IMPORTANT: If you are working in a non-DaVinci Resolve Color managed workflow and activate the HDR (ST.2084/HLG) scopes, the waveform will always reflect HDR levels, even if you've manually selected the output color space as Rec.709, and thus not be 100% accurate.

Changing Vectorscope Scales

A Vectorscope Scale Style submenu in the Video Scopes option menu (the three dots menu) lets you choose how you want the hue reference indicators to be displayed.



Vectorscope scale options

There are four available graticule options:

Off: Disables all graticule overlays in the Vectorscope, giving you a clean view of the Vectorscope graph against black.

Standard: The default layout. Cross hairs indicate the center of the scope, while boxes indicate the center target of each primary and secondary hue (red, magenta, blue, cyan, green, yellow). An outer circle provides a general frame of reference for the angle that any part of the graph might fall along.

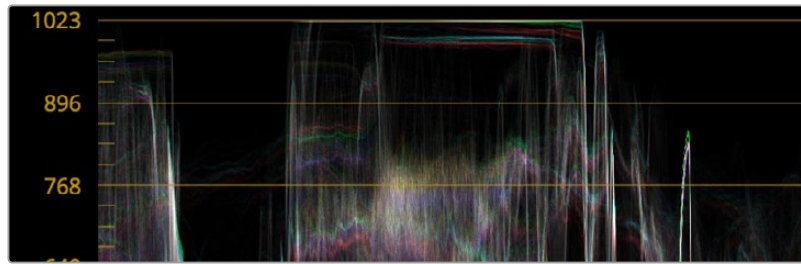
Simplified: Shows a crosshairs to indicate the center of the scope and smaller crosshairs to indicate each primary and secondary hue. Useful when you want references without extra complications.

Hue Vectors: A graticule designed to provide a more useful colorist reference for creative decisions and image comparison. This isn't simply decorative. Graticule lines stretch along the reference angle of each primary and secondary hue from the hue targets to the center, providing a more immediately useful frame of reference when comparing different vector scope graphs to one another. These reference lines fade out within the region of average saturation in most images, so they don't get in the way of discerning faint details. The center crosshairs are aligned with the red-cyan and yellow-blue axes, indicating the naturalistic warm to cool axis that lies between them. Tic marks indicate both the 75% and 100% levels of image saturation for each hue.

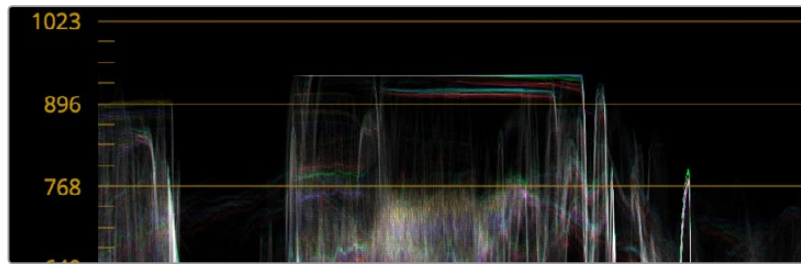
You also have the option of choosing whether or not the primary and secondary hue targets are shown for 75% image saturation, 100% image saturation, or both, and whether or not you want to show the text labels for each hue.

Displaying Scopes With Data or Video Levels

For projects being worked on at video levels, a setting in the Video Scope option menu's Waveform Scale Style submenu lets you toggle between displaying video scopes scaled to either Data Levels (the default) or Video Levels (by turning on Video Level Scopes). This only affects how your scopes are displayed; it has no effect on monitored or rendered output.



The Waveform scope shown at default Data Levels,



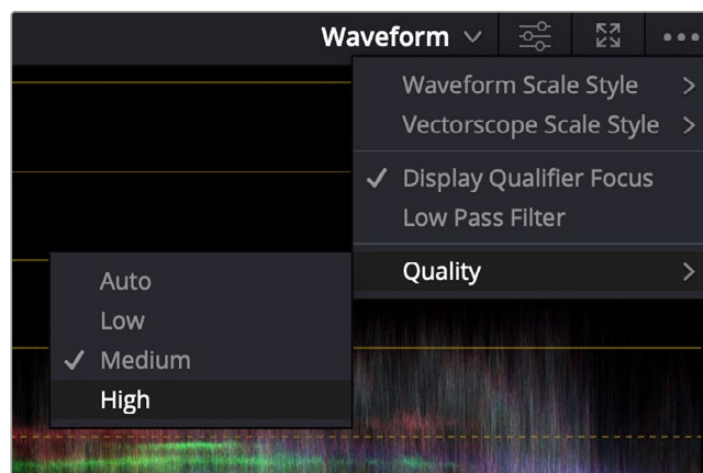
The same waveform shown with Video Level Scopes turned on

Video Scope Performance and Detail

All video scopes were updated for DaVinci Resolve 16 to show more detail with faster performance than in previous versions. While they're GPU-accelerated, the video scopes require a certain amount of video processing power to operate. Depending on which combination of video scopes and scope options you have selected, you may notice your real time processing capabilities ever so slightly affected. Closing the video scopes frees up all processing for color correction and effects. On high-powered workstations, you probably won't notice a difference, but on less powerful computers, closing the scopes might make a small difference.

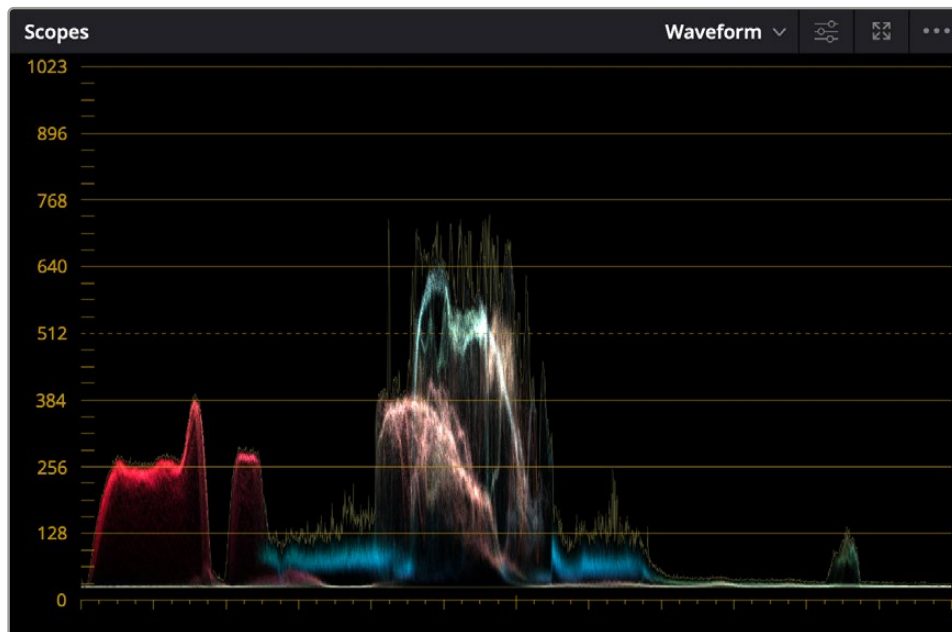
Two global settings in the Video Scope option menu affect the performance and detail of all video scopes equally.

- On slower workstations, a quality submenu lets you choose High, Medium, or Low quality scope drawing, to trade off legibility for performance. High quality shows you the most information about the video signal, while an Auto option makes a selection based on your workstation's capabilities.

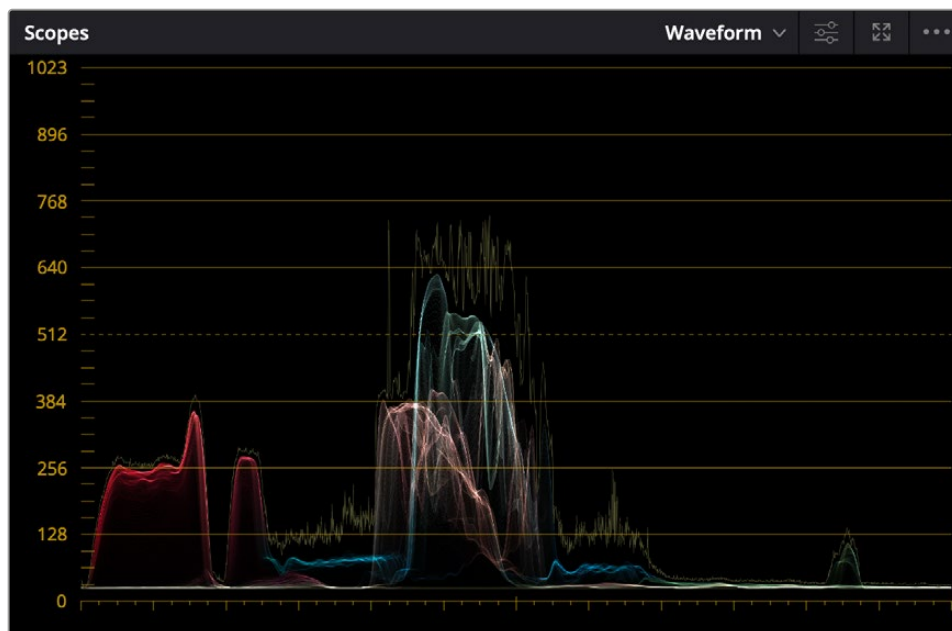


The Quality setting in the Video Scope option menu

- All video scopes have a Low Pass Filter setting in the Video Scope option menu that filters out a signal's noise to make the scope graphs clearer to read. While this can act as an “x-ray” to better see detail in the scope graph interiors, it can make the highlight and shadow excursions on the graph seem a bit shy of where they actually are, so it's recommended to enable the “Extents” option of whichever scope you're using to get an unambiguous look at the maximum excursions in your scopes. Extents draws an outline highlighting all graph excursions to show you the true level of all overshoots and undershoots in the video signal, even when the Low Pass Filter is on.



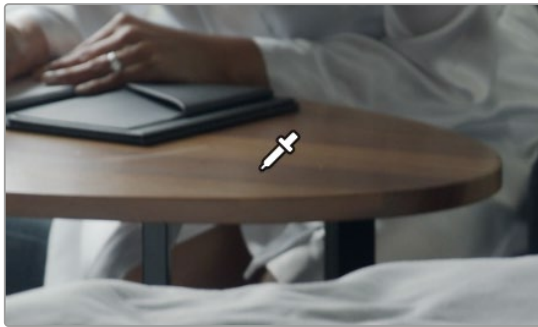
The Waveform scope with Low Pass Filter turned off



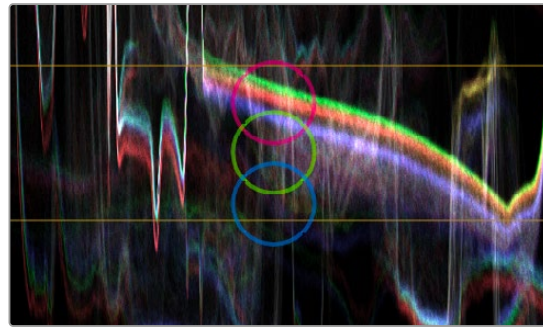
The Waveform scope with Low Pass Filter turned on

Display Qualifier Focus in Video Scope Graphs

The Display Qualifier Focus setting in the Video Scope option menu helps you identify which features in the video image correspond to which parts of the video scopes. With Display Qualifier Focus turned on, choosing the Qualifier mode of the Viewer and moving the eyedropper around the image draws circles around the sampled pixels as they appear in the currently visible video scope graph. If multiple video scopes are visible, each scope will have an indication of the location of the sampled pixels that's specific to each scope.



Hovering the eyedropper over a feature in the Viewer



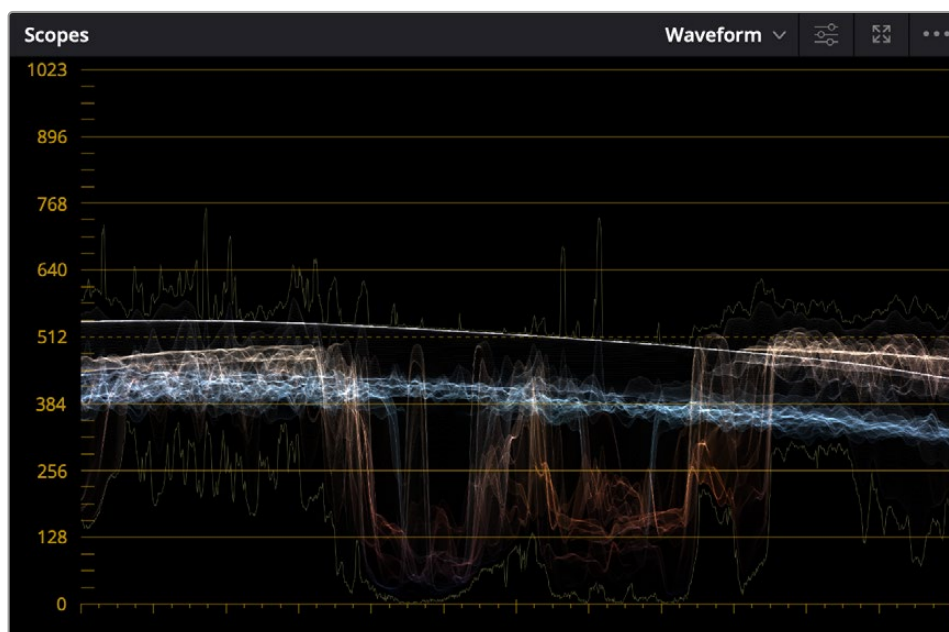
The regional analysis of those pixels shown in the overlaid red, blue, and green waveforms of the Waveform scope.

Explanation of Each Video Scope

There are five available video scopes, each of which shows a different analysis of the video signal you're adjusting.

Waveform Monitor

Overlays waveform analyses of the Y (luma/luminance), CBCR (the color difference channels of Y'CBCR), or RGB (red, green, and blue) channels over one another so that you can see how they align.



The Waveform scope showing Y' (luma) only, with colorization and extents enabled

The Y option presents a true Luma scope that can have Colorize enabled to show false color, which lets you see which colors in the Viewer image are where in the video scope graph.

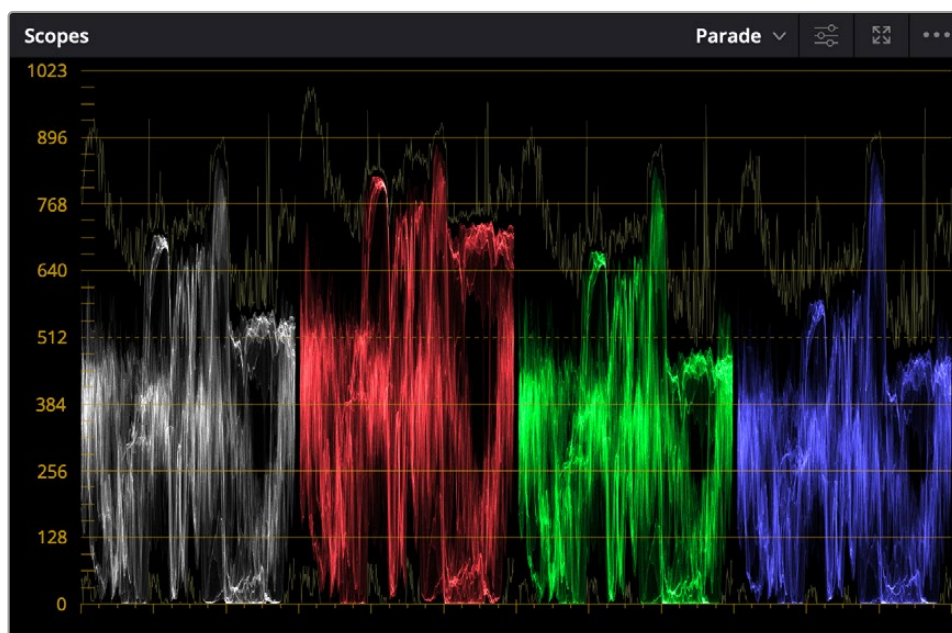
For the RGB scopes, the relative heights of the red, green, and blue graphs indicate is identical to the description seen for the Parade scope below, and with color enabled the red, green, and blue graphs are tinted with the color they represent. This makes it easy to see where all three graphs align, by looking for where parts of the waveform monitor appear in white, which is the result of the red, green, and blue graphs lining up and adding their color together.

Parade

The Parade scope shows separate waveforms side by side that analyze the strength of individual video signal components. The Parade scope can be set to analyze RGB, YRGB, and Y'CBCR.

By showing a comparison of the intensity of the luma, red, green, and blue channels, the Parade scope makes it possible to detect and compare imbalances by comparing the relative heights of the RGB graphs in the highlights (the top of the R, G, and B graphs), shadows (the bottom of the R, G, and B graphs), and midtones (the middle of the R, G, and B graphs) for the purposes of identifying color casts and performing scene-by-scene correction.

When the YRGB channels are taken altogether, the bottom of all graphs indicates the black point of the image, while the tops of all graphs indicate the white point. It then follows that the difference between the height of the bottom and top of all graphs indicates the overall contrast ratio of the image you're evaluating. Tall parade graphs indicate a wide contrast ratio, while short parade graphs indicate a narrow contrast ratio.



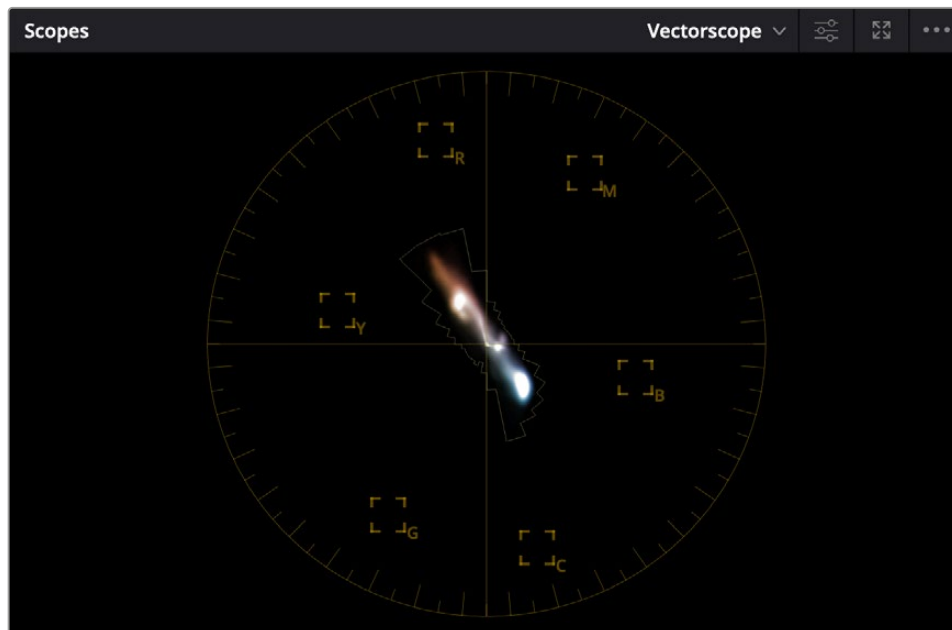
The Parade scope showing YRGB waveforms, with colorization and extents enabled

Vectorscope

Measures the overall range of hue and saturation within an image. Measurements are relative to a centered graticule that you can enable that's overlaid on the scope, which provides a frame of reference via crosshairs. DaVinci Resolve has a traditional vectorscope, the graph of which emulates a trace-drawn graph, with 75 percent color bar targets indicating the angle of each of the primary and secondary colors around the edge of the graph, and an optional skin tone reference graticule (otherwise known as the In-phase reference).

The Vectorscope can have Colorize enabled to show false color which lets you see which colors in the Viewer image are where in the video scope graph.

More saturated colors in the frame stretch those parts of the graph farther toward the edge, while less saturated colors remain closer to the center of the vectorscope, which represents 0 saturation. By judging how many parts of the vectorscope graph stick out at different angles, you can see how many hues there are in the image, with the specific angle of each part of the graph showing you which hues they are. Furthermore, by judging how well centered the middle of the vectorscope graph is relative to the crosshairs, you can get an idea of whether there's a color imbalance in the image. If the vectorscope graph is off-centered, the direction in which it leans lets you know that there's a color cast (tint) in the image.



The Vectorscope shown in 2x mode, with combined highlights, mid-tones, and shadows extents

Histogram (RGB/YRGB parade histogram)

Displays a statistical analysis of how many pixels of each color channel lie at each percentage of tonality, plotted along a digital scale from 0 percent (black) to 100 percent (white). Comparing the left, middle, and right parts of the Y, R, G, and B graphs (Y is optional) lets you evaluate the color balance in the shadows, midtones, and highlights of the image.



The Histogram scope set to YRGB

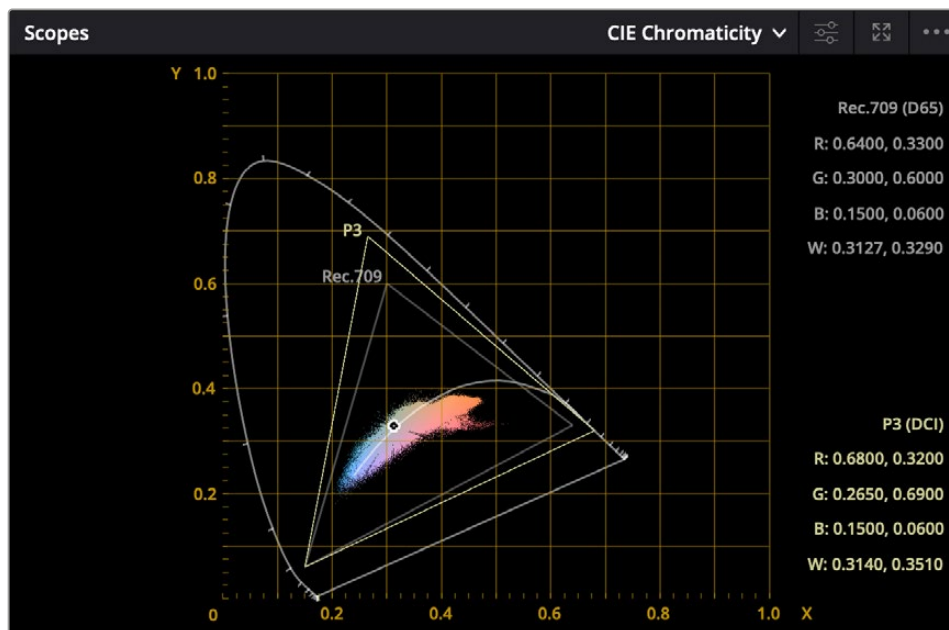
Taken altogether, the left of all graphs indicates the black point of the image, while the right of all graphs indicate the white point. It then follows that the difference between the width of the left and right of all graphs indicates the overall contrast ratio of the image you're evaluating. Wide histogram graphs indicate a wide contrast ratio, while narrow histogram graphs indicate a narrow contrast ratio.

CIE Chromaticity Scope

DaVinci Resolve contains CIE 1931 xy and CIE 1976 uv scopes, which let you see the current image analysis as a graph superimposed against a triangle that represents the tristimulus values of the color gamut you're working within, along with an indication of the current white point. A label shows the currently selected gamut, with the specific coordinates of these Red, Green, and Blue values as well as the White point, while the overall "horseshoe" shape represents the entire range of visible light, all plotted against an xy graph.

The white point ordinarily appears atop a curve along the center of the shape. This curve indicates the black body locus, along which the various color temperatures of an image's white point correlate, from orange-ish warm to blue-ish cool. This black body locus corresponds to the color temperatures obtained by progressively heating carbon to different temperatures.

You also have the option of showing a second gamut triangle, in cases where you want to compare the current analysis to two different gamuts. This can be useful when you need to create deliverables in multiple gamuts, and you want to see which parts of the video signal are safe in all gamuts and which parts are exceeding the smaller of the two. When you do this, both gamuts are labeled, and the coordinates of the Red, Green, and Blue values of both gamuts are listed to the side.



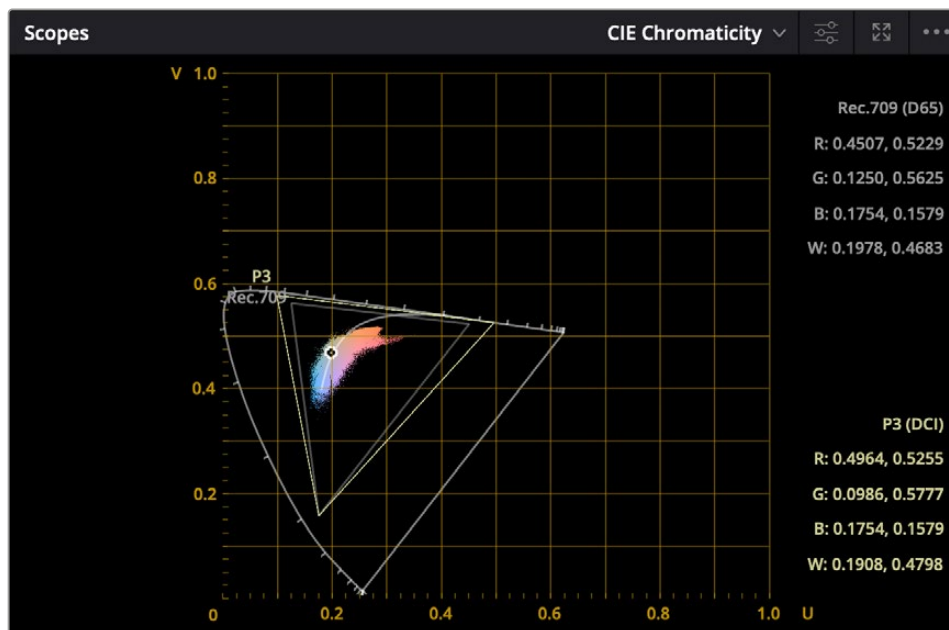
The CIE 1931 xy scope showing both P3 and Rec. 709 gamuts

In a way, the Chromaticity scope is a 3D scope, but it's drawn as if you're looking down at the top of a 3D shape that plots every color value in an image in X, Y, Z space, but you can only perceive the 2D outline around the widest parts of this shape drawn on an X, Y plot. The graph does indeed represent every single value found within a 3D plot of the image data, but the triangle only indicates the widest "slice" of the current gamut and of this 3D shape within the mid-tones.

What this means is that while the Chromaticity scope's graph gives you a rough idea of whether or not the current image is within gamut relative to your delivery spec, it's not exact and it's not foolproof, as part of the image data could fall within this wide triangle and yet overshoot the required gamut elsewhere in the highlights at the top or the shadows near the bottom of the 3D shape you're looking down on. On the other hand, if any part of the graph does extend past the boundaries of the gamut triangle, then that definitely indicates a gamut violation.

You can add an additional gamut triangle in the scope settings in order to compare the color space you're working in to another color space, for reference.

Most people who've had any exposure to color grading concepts are familiar with the traditional CIE 1931 horseshoe graph, which plots the portion of the spectrum visible to the human eye according to studies done in the late 1920s (subsequent studies have confirmed this analysis). The optional CIE 1976 uv graph is based on an updated color space (CIELUV) that was an attempt by the International Commission on Illumination (CIE) to create a more perceptually uniform graph of color. Whereas the CIE 1931 analysis visually exaggerates certain parts of a color graph, the CIE 1976 graph draws all parts of a color graph more or less proportionally to one another. Overall, neither analysis is more "correct" than the other, it's simply a matter of what you prefer to look at.



The CIE 1976 scope showing both P3 and Rec. 709 gamuts

Panning and Zooming the Video Scopes

If you want to examine any part of a video scope's graph in more detail, you can do the following:

To zoom into a graph: Hold the Option key down, and roll the scroll control.

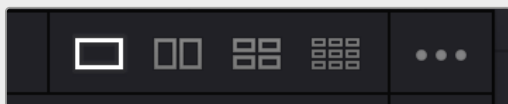
To pan around a graph: Click and drag with the middle pointer button.

Customizing the Video Scopes

Once you've opened the video scopes, there are a variety of ways you can customize how the scopes look, and expose additional onscreen graticule information to help you measure what you need.

Methods of customizing the Scopes window:

- **To change the size of the Scopes window:** Drag the lower right-hand corner to resize the Scopes window to the desired size.
- **To change how many scopes are displayed at once:** Click one of the buttons in the upper right-hand corner of the Scopes window to set 1, 2, 4, or 9 up arrangements of the video scopes. You can also choose how many scopes are simultaneously displayed by choosing Workspace > Video Scopes > 1-Up, 2-Up, or 4-Up.
- **To change which scopes appear in which pane:** Click the Name drop-down at the upper left corner of each scope's pane, and choose a different scope. If you like, you can instantiate more than one of each kind of video scope in multi-scope layouts, for instances where you want to view variations on a particular type of scope with different settings. For example, you might want to expose three Vectorscopes, setting each to a different tonal range so you can simultaneously view Vectorscope analyses of Low (shadows), Mid (midtones), and High (highlights).

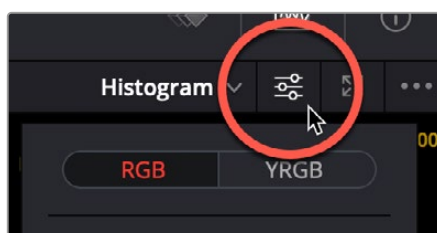


Different layout options in the Scope window

Once open, you can resize the Scopes window to make it as large or small as you require, positioning on a second display if you want to make it even larger.

To customize each video scope's display options:

- 1 Click the Option button to the right of the Scope drop-down menu to expose the current scope's Custom Controls window.



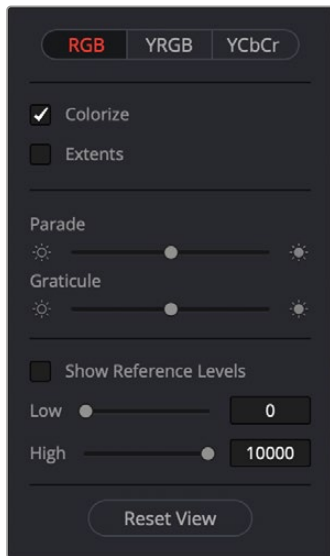
Clicking to open the current scope's custom controls

- 2 Adjust any of the available controls to customize the look of that particular scope.
- 3 Click anywhere outside of the Custom Controls window to make it disappear.

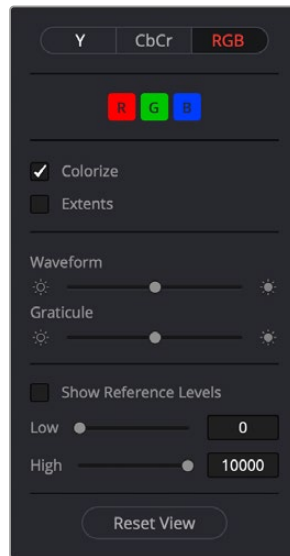
Parade Scope Display Options

The Parade scope has the following options:

- RGB, YRGB, and YCbCr modes, allowing you to evaluate more channels than before.
- A colorize checkbox lets you view graphs in monochrome or false-color (indicating red, green, and blue).
- An Extents checkbox draws an outline highlighting all graph excursions, to unambiguously show you all overshoots and undershoots in each waveform.
- The Parade slider makes that scope's graph brighter or dimmer. Brighter graphs make it easier to see fine detail, but harder to see which parts of the graph are stronger and weaker.
- The Graticule slider makes that scope's scale brighter or dimmer, making it more or less visible (or distracting) relative to the graph.
- The Show Reference Levels checkbox lets you turn on adjustable Low and High reference level markers by setting the low and high sliders to something other than their defaults. This is especially useful for HDR grading where you're working within a specific peak luminance threshold.



Options in the Parade scope



Options in the Waveform scope

Waveform Scope Display Options

The Waveform scope has the following options:

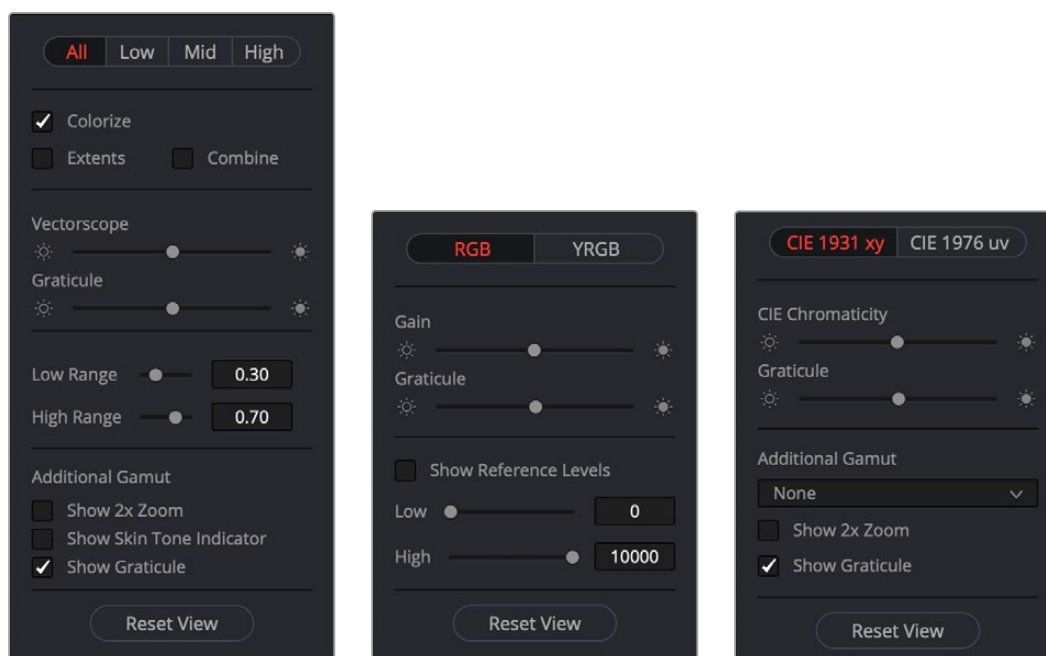
- Y (luma) and CbCr (chrominance) modes to show you the true luma or chroma signals in isolation, and RGB to show you only an RGB analysis.
- In RGB mode, R, G, and B buttons that can be individually toggled on and off to see any combination of graphs.
- A colorize checkbox lets you view overlaid graphs in monochrome or false-color (indicating red, green, and blue). If you're only enabling the Y or C scopes, the different areas of these graphs are drawn with color taken from the image being analyzed, which makes it easier to see which part of the scope graph corresponds to which part of the image.
- An Extents checkbox draws an outline highlighting all graph excursions, to unambiguously show you all overshoots and undershoots in each waveform.
- The Waveform slider makes that scope's graph brighter or dimmer. Brighter graphs make it easier to see fine detail, but harder to see which parts of the graph are stronger and weaker.
- The Graticule slider makes that scope's scale brighter or dimmer, making it more or less visible (or distracting) relative to the graph.
- The Show Reference Levels checkbox lets you turn on adjustable Low and High reference level markers by setting the low and high sliders to something other than their defaults. This is especially useful for HDR grading where you're working within a specific peak luminance threshold.

Vectorscope Display Options

The Vectorscope has the following options:

- Choose ALL to view the entire tonal range of the image as a Vectorscope graph analysis, or to selectively view just the shadows (Low), mid-tones (Mid), or highlights (High) of the image as a vectorscope graph analysis.
- A Colorize checkbox draws the different areas of these graphs with color taken from the image being analyzed, which makes it easier to see which part of the scope graph corresponds to which part of the image. With colorize turned off, graphs only appear white.

- An Extents checkbox draws an outline highlighting all graph excursions to unambiguously show you all overshoots and undershoots.
- A Combine checkbox lets you see simultaneous overlapping extents for the highlights, mid-tones, and shadows overlaid on one another.
- The Vectorscope slider makes that scope's graph brighter or dimmer. Brighter graphs make it easier to see fine detail, but harder to see which parts of the graph are stronger and weaker.
- The Graticule slider makes that scope's scale brighter or dimmer, making it more or less visible (or distracting) relative to the graph.
- Low Range and High Range sliders let you manually define the boundaries separating shadows from mid-tones and highlights. Low Range defaults to 0.30, and High Range defaults to 0.70.
- Show 2X Zoom zooms the vectorscope graph by 200%, making it easier to see fine detail and use the vectorscope with charts.
- Show Skin Tone Indicator checkbox shows a line at the traditional Inphase angle that is useful as a general guidepost for average skin tone hue.
- Show Graticule checkbox lets you show or hide the circular degrees indicator surrounding the outer edge and the cross hairs that indicate the center of the vectorscope.



Options in the Vectorscope

Options in the Histogram scope

Options in the Chromaticity scope

Histogram Display Options

The Histogram has the following options:

- You can view either RGB or YRGB histograms.
- Gain slider scales that scope's graph taller or shorter. Taller graphs expand to show greater detail in the histogram's peaks, shorter graphs reduce the apparent detail.
- Graticule slider makes that scope's scale brighter or dimmer, making it more or less visible (or distracting) relative to the graph.
- The Show Reference Levels checkbox lets you enable adjustable Low and High reference level markers by setting the Low and High sliders to something other than their defaults. These reference markers are especially useful for HDR grading where you're working within a specific peak luminance threshold.

CIE Chromaticity Display Options

The CIE Chromaticity scope has the following options:

- You can view a chromaticity analysis in either a CIE 1931 xy graph, or a CIE 1976 uv graph.
- CIE Chromaticity slider makes that scope's graph brighter or dimmer. Brighter graphs make it easier to see fine detail, but harder to see which parts of the graph are stronger and weaker.
- Graticule slider makes that scope's scale brighter or dimmer, making it more or less visible (or distracting) relative to the graph.
- Additional Gamut drop-down menu lets you expose a second gamut triangle, for instances where you want to compare how an image fits into two different gamut ranges.
- Show 2X Zoom zooms the CIE graph and graticule by 200%, making it easier to see fine detail and use the vectorscope with the charts.
- Show Graticule checkbox lets you show or hide the circular degrees indicator surrounding the outer edge and the cross hairs that indicate the center of the vectorscope.